Can an MLP Absorb Its Skip Connection Exactly?

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Abstract

Residual networks add each block's input to its output; skipless ones now train too. Can a block without biases absorb its skip: can a skipless block of the same kind compute the same function? Here we show that almost every ReLU or GELU block at least as wide as the model dimension d_{model} absorbs its skip with exactly d_{model} more units. Squared-ReLU, gated (ReGLU, SwiGLU, GeGLU), $\tanh$ and attention blocks never do, however wide. Without normalization, no residual squared-ReLU, gated or $\tanh$ network equals a skipless one. Almost no residual ReLU or GELU network equals a skipless one whose first block has fewer than $2d_{\text{model}}$ units. Under RMSNorm, almost no residual network equals any skipless one. Yet without normalization, every skipless network is a limit of residual ones. These results say when a residual network can be rewritten exactly without skips, and at what cost.

1 Introduction

Most deep networks trained today add the input of each block to its output (He et al., 2016; Vaswani et al., 2017). The usual reason for these skip connections is training: the sum gives the gradient a short path through the network, and it makes the identity map easy to express (Hardt and Ma, 2017; Veit et al., 2016).

Skipless networks now train too. He and Hofmann (2024) remove the skips of transformer blocks without losing training speed, and Ji et al. (2026) train a skipless ViT-B on ImageNet that outperforms the residual one. So does a skip only help training, or does it also let a block compute functions that it cannot compute without one?

We answer this question exactly. Take a block B without biases that acts on tokens of dimension d_{model}: an MLP block, or an attention block. Its *width* is its number of hidden units, or of heads. We say that B *absorbs* its skip if the residual block $x \mapsto x + B(x)$ computes the same function as a skipless block of the same kind that is at least as wide. Blocks of the same kind are both attention blocks, or both MLP blocks of the same type and activation (Section 2). If B absorbs its skip, a trained residual network can be rewritten without that skip, with the same outputs at every input. The skip is *worth* n units if B absorbs it with n more units and with no fewer; if no width suffices, it is worth more than any number of units.

One block: the answer depends on the activation.

- For almost every ReLU or GELU block with at least d_{model} hidden units, the skip is worth exactly d_{model} units (Theorem 3.1(ii)). Each new unit has the input weights of an existing unit with the opposite sign. For ReLU, such sign flips are known from two-layer networks with one output and a linear term (Petzka et al., 2020).
- For squared-ReLU (ReLU^2), gated (ReGLU, SwiGLU, GeGLU) and $\tanh$ blocks, no width suffices (Theorem 3.3).

Author contributions. Antonij Mijoski (AM) proved the absorption result for a single ReLU block (Theorem 3.1(i), Theorem A.4), the difficult case, paving the way forward. Marko Karbevski (MK) conceived and led the study. MK extended the tools to GELU (Theorem A.7), proved the cases of ReLU^2 and of the gated variants ReGLU, SwiGLU and GeGLU (Theorems 3.3 and 4.1) and the results at depth (Theorem 4.5 and Proposition 4.4). Both authors worked on the writing. All of MK's work on the results and the proofs was done with AI assistants.

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- For attention blocks, no number of heads suffices (Theorem 3.4).

These answers are for blocks without biases, as in many current language models (Team OLMo et al., 2025; Yang et al., 2025). With biases, every ReLU^2 and gated block does absorb its skip, with more units. So does an attention block, but only with rotary embeddings (Su et al., 2021). A $\tanh$ block still never does (Section 2, Appendix C).

Why the other blocks never absorb. Follow the input along a ray, $x = tu$ with $t > 0$. The skip adds tu, which grows like t. A ReLU^2 block grows like t^2. A gated block is of order t^2 near the origin, because its gate vanishes at 0. A $\tanh$ block stays bounded. So none of them can produce the term tu (Figure 1, Section 3.2). For attention the reason is a zero token: every head gives it the average of the tokens it attends to, and a skip gives it nothing (Section 3.3).

Networks: the negative answers carry over. A network has a linear embedding E before its blocks and a linear readout D after them. With skips, an input X also reaches the output directly, as XED, and we assume $ED \neq 0$. In a pre-RMSNorm network, every block and the readout read their input through RMSNorm, regularized by a constant $\varepsilon \geq 0$.

- Without normalization, no residual network of ReLU^2, gated or $\tanh$ blocks equals a skipless network, whatever its depth and widths (Theorem 4.1). For ReLU^2 and gated blocks, the same holds under pre-RMSNorm with $\varepsilon > 0$.
- No residual network of attention blocks equals a skipless one of any depth when the attention has no mask. With a causal mask, the same holds once a sequence has at least two more tokens than the residual network has blocks (Theorem 4.2).
- Under pre-RMSNorm the kind of block no longer matters: almost no residual network equals a skipless network, whatever its depth and widths (Theorem 4.3). A skip keeps the norm of each token, and a skipless network loses it at its first normalization. This covers pre-RMSNorm transformers.
- ReLU and GELU networks without normalization are the exception. Almost every residual one equals a skipless network with d_{model} more units per block (Proposition 4.4). Almost none equals a skipless network whose first block has fewer than $2d_{\text{model}}$ units in all (Theorem 4.5). When the residual blocks have $2d_{\text{model}}$ units or more, as in the usual width $4d_{\text{model}}$, the number of extra units that are needed is known only in part (Section 4.4).

Skipless networks as limits. Without normalization, every skipless network is a limit of residual networks of the same widths, uniformly on bounded sets of inputs (Proposition 5.1(i)). So where no residual network equals a skipless one, residual networks still come arbitrarily close to every skipless network. How closely a skipless block can approximate a residual one is another question. We treat it only for attention blocks, where a residual block stays at a fixed distance from every skipless one (Theorem 3.4(ii)).

Table 1 collects the results, and the proofs are in the appendix.

2 Setup

One equation states the question: can a block absorb a linear term?

Blocks. A hidden state is a matrix $h \in \mathbb{R}^{T \times d_{\text{model}}}$ whose rows are the T tokens of a sequence. An MLP block acts on each token separately. It has two weight matrices, a read W_{read} and a write W_{write}, and a fixed map f between them. Its parameters are $\theta = (W_{\text{read}}, W_{\text{write}})$. On a row $x \in \mathbb{R}^{1 \times d_{\text{model}}}$ of a hidden state it computes

$$B_\theta(x) = f(xW_{\text{read}})\, W_{\text{write}}. \tag{1}$$

In a vanilla block with m hidden units, $W_{\text{read}} \in \mathbb{R}^{d_{\text{model}} \times m}$, $W_{\text{write}} \in \mathbb{R}^{m \times d_{\text{model}}}$, and f applies an activation σ to each unit: ReLU, GELU, $\tanh$, or the square of ReLU, ReLU^2. In a gated block the read has two halves, $W_{\text{read}} = [\, W_{\text{g}} \mid W_{\text{v}} \,]$ with $W_{\text{g}}, W_{\text{v}} \in \mathbb{R}^{d_{\text{model}} \times m}$. Column i of W_{g} is the gate read of unit i, and column i of W_{v} is its value read. The unit multiplies $\sigma\big((xW_{\text{g}})_i\big)$ by its value $(xW_{\text{v}})_i$. The activation σ of a gated block is called its gate: ReLU in ReGLU, SiLU in SwiGLU and GELU in GeGLU (Shazeer, 2020). Two MLP blocks are of the same kind if they are of the same

Proven	**ReLU**	**GELU**	**tanh**	**ReLU^2, ReGLU**	**SwiGLU, GeGLU**	**Attention**
Can a skipless network equal a residual one? Without normalization:						
one block, same width	measure zero Thm. 3.1(i)		never Thm. 3.3			never Thm. 3.4
one block, more hidden units or heads	exactly d_{model} more Thm. 3.1(ii)		never Thm. 3.3			never Thm. 3.4
one block with embedding and readout, same width	measure zero Prop. 3.2		never Thm. 4.1(ii)	never Thm. 4.1(i)		never Thm. A.13(iii)
deeper networks, same depth and widths	measure zero if the first block has fewer than $2d_{\text{model}}$ units Thm. 4.5, and within an open set of weights Prop. 4.7; else unknown	measure zero companion	never Thm. 4.1(ii)	never Thm. 4.1(i)		never with no mask or more tokens than blocks, else measure zero Thm. 4.2
networks, same depth, more hidden units or heads per block	for almost all weights, d_{model} more per block suffice Prop. 4.4 and the first block needs $2d_{\text{model}}$ units Thm. 4.5; every block needs $2d_{\text{model}}$ units if all residual blocks are injective with ReLU as activation Prop. 4.6		never Thm. 4.1(ii)	never Thm. 4.1(i)		never with no mask or more tokens than blocks, else measure zero Thm. 4.2
With pre-RMSNorm:						
networks, $\varepsilon > 0$	measure zero, at any width Thm. 4.3			never Thm. 4.1(i)		never with no mask or at least two more tokens than blocks Thm. 4.2; else measure zero Thm. 4.3
networks, $\varepsilon = 0$	measure zero, at any width Thm. 4.3					
Is every skipless network a limit of residual ones of the same widths?						
without normalization	yes Prop. 5.1(i)					
with pre-RMSNorm, $\varepsilon = 0$	yes Prop. 5.1(ii)					
with pre-RMSNorm, $\varepsilon > 0$	unknown					

Table 1: Never: for no weights of the residual network. Measure zero: only for a null set of its weights, so with probability zero for random weights. Exactly d_{model} more: for almost every block, d_{model} more hidden units suffice and fewer do not. Companion: Karbevski and Mijoski (2026). No biases. Scope in Section 2.

type, vanilla or gated, and have the same activation. All attention blocks are of one kind; they follow in Section 3.3.

Absorbing a linear term. A skip adds x to the output of a block. More generally, can a block absorb a linear term xM, for a $d_{\text{model}} \times d_{\text{model}}$ matrix $M \neq 0$? We look for parameters θ' of a block of the same kind, and of any width, such that

$$xM + B_\theta(x) = B_{\theta'}(x) \qquad \text{for all } x \in \mathbb{R}^{1 \times d_{\text{model}}}. \tag{2}$$

The skip is $M = I$; other M cover learned linear skips and networks of one block (Section 3.1).

Networks. A network has a linear embedding $E \in \mathbb{R}^{d_{\text{in}} \times d_{\text{model}}}$, then $L \geq 1$ blocks $B_1, \ldots, B_L$, each with a skip A_ℓ, then a linear readout, or unembedding, $D \in \mathbb{R}^{d_{\text{model}} \times d_{\text{out}}}$, and an output function C. It maps an input $X \in \mathbb{R}^{T \times d_{\text{in}}}$ to

$$\mathcal{M}(X) = C \circ D \circ (A_L + B_L) \circ \cdots \circ (A_1 + B_1) \circ E\,(X),$$

where E and D stand for the maps $X \mapsto XE$ and $h \mapsto hD$. The skip A_ℓ of block ℓ is the zero map in a *skipless* network and the identity map I in a *residual* one. So the hidden states are $h_0 = XE$ and $h_\ell = B_\ell(h_{\ell-1})$ in a skipless network and $h_\ell = h_{\ell-1} + B_\ell(h_{\ell-1})$ in a residual one, and the readout

gives $y = h_L D$. The output function C is set by the task: the identity for regression, the logistic sigmoid of each output for multilabel classification, and the softmax for multiclass classification (Bishop, 2006, Sections 5.1 and 5.2). The network returns $\mathcal{M}(X) = C(y)$. The function C has no weights, and we ignore it: equality of networks, and every result below, concern y, the logits of a classifier. Networks with the same y have the same $\mathcal{M}$, but not conversely; a softmax, for instance, does not change when the same number is added to every output.

Normalization. RMSNorm, written N, divides each row v of a hidden state by $\sqrt{\|v\|^2/d_{\text{model}} + \varepsilon}$, with a constant $\varepsilon \geq 0$. In a *pre-RMSNorm* network every block and the readout read the normalized state $N(h)$, while the skips carry h itself. So a skipless network computes $h_\ell = B_\ell\big(N(h_{\ell-1})\big)$, a residual one $h_\ell = h_{\ell-1} + B_\ell\big(N(h_{\ell-1})\big)$, and both read out $y = N(h_L)D$. For $\varepsilon = 0$, N is defined only on hidden states without zero rows. A network is then defined at the inputs at which no state that it normalizes has a zero row. Two such networks are equal if they agree on a dense set of inputs at which both are defined.

Biases. Blocks have no biases, and the answers depend on it (Appendix C). With biases, every ReLU, GELU, ReLU^2 and gated block absorbs its skip with more units; for almost every ReLU, GELU and gated block, with exactly d_{model} more. For gated blocks the reason is short. A unit whose gate read is zero multiplies its value by a constant, so it is affine in x, and a block absorbs its skip with d_{model} such units. An attention block with biases absorbs its skip only with rotary embeddings: heads whose queries and keys are their biases attend by position alone, and such heads add up to a skip. No $\tanh$ block absorbs its skip, with or without biases.

Scope. Every MLP block has at least d_{model} hidden units, so that its read can have rank d_{model}. Residual networks have $ED \neq 0$. For simplicity, all MLP blocks of a network are of one kind, all attention blocks of a network have the same number of heads, and $d_{\text{in}}, d_{\text{out}} \geq d_{\text{model}} \geq 2$. Most statements take one kind of block. Theorem 4.3 allows blocks of any kind, and so covers networks that alternate attention and MLP blocks, as transformers do. The normalization is RMSNorm, without a learned gain, and the embedding is linear, without positional terms; we do not treat LayerNorm. The appendix states each result in greater generality. A skipless network that is compared with a residual one may have another hidden dimension, unless the statement says "the same d_{model}".

Generic weights. A statement holds for *generic* weights, or for almost every choice of weights, if it holds outside a set of weights of Lebesgue measure zero.

3 One block

For almost every ReLU or GELU block, a skip is worth exactly d_{model} hidden units. For ReLU^2, gated, $\tanh$ and attention blocks it is worth more than any number.

3.1 ReLU and GELU blocks: a skip is worth d_{model} units

ReLU and GELU both satisfy

$$\sigma(z) - \sigma(-z) = z. \tag{3}$$

Let unit i have read w_i, a column of W_{read}, and write $\hat{w}_i$, a row of W_{write}. By (3), unit i and its mirror image, a unit with read $-w_i$ and write $-\hat{w}_i$, add up to the linear term $(xw_i)\,\hat{w}_i$. This gives two ways to absorb xM.

With more units. For a row $\gamma \in \mathbb{R}^{1\times d_{\text{model}}}$, a new unit with read $-w_i$ and write $-\gamma$, with the write of unit i changed to $\hat{w}_i + \gamma$, adds $(xw_i)\,\gamma$ to the block. So each new unit adds one rank-one term, and d_{model} new units, mirroring d_{model} units whose reads span $\mathbb{R}^{d_{\text{model}}}$, add any xM.

At the same width. Flipping the read of unit i to $-w_i$ changes the block by $-(xw_i)\,\hat{w}_i$. So B_θ absorbs xM at the same width if some set S of units has $W_{\text{read}}[:, S]\, W_{\text{write}}[S, :] = -M$, where $W_{\text{read}}[:, S]$ and $W_{\text{write}}[S, :]$ keep the columns and rows of the units in S.

The next theorem says that at the same width these flips are the only way, and that fewer than d_{model} new units do not suffice. A block is *nondegenerate* if the columns of W_{read} are nonzero and pairwise non-collinear and the rows of W_{write} are nonzero.

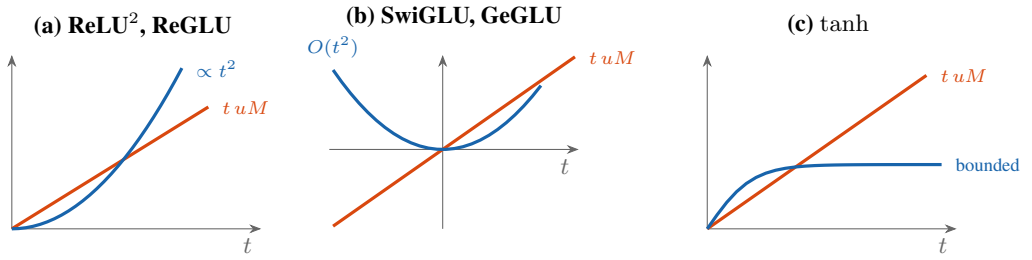


Figure 1: Why no width of ReLU^2, gated or tanh units absorbs the skip. Along a ray $x = tu$ the skip grows like t, and the difference of two blocks cannot: (a) it grows like t^2; (b) it is of order t^2 near $t = 0$; (c) it stays bounded.

Theorem 3.1 (Absorption with d_{model} more units)**.** *Let σ be ReLU or GELU, and fix a matrix $M \neq 0$.*
(i) *If B_θ is nondegenerate, some block $B_{\theta'}$ of the same width satisfies (2) if and only if some set S of hidden units has $W_{\text{read}}[:, S]\, W_{\text{write}}[S, :] = -M$. For generic θ, no block of the same width does.*
(ii) *If M is invertible and W_{read} has rank d_{model}, some block $B_{\theta'}$ with d_{model} more hidden units satisfies (2), and for generic θ none with fewer does.*

Proof in Appendix A.1.

In (i), the equation $W_{\text{read}}[:, S]\, W_{\text{write}}[S, :] = -M$ is polynomial in the weights for each of the finitely many sets S, and it fails when the writes vanish, so it holds only on a set of weights of measure zero. By (ii), for almost every block the skip, $M = I$, is worth exactly d_{model} units.

The proof matches the units of the two blocks. The even part of the activation, $\big(\sigma(z)+\sigma(-z)\big)/2$, matches each unit of $B_{\theta'}$ to a unit of B_θ, up to sign and scale: for ReLU through the kinks of $|z|$, and for GELU through the linear independence of ridge functions, which depend on x only through one product xw. The odd part, $z/2$, turns each sign flip into a linear term. Each extra unit then adds at most one rank-one term, and $M = I$ has rank d_{model}. For two-layer ReLU networks with one output, with biases and a linear term, Petzka et al. (2020, Theorem 1) show that permutations, rescalings and such sign flips are the only transformations between two irreducible networks that compute the same function; part (i) is the analogue for blocks with several outputs and without biases, and for GELU. Identity (3) alone does not make these flips the only way: Remark A.8 gives an activation that satisfies it and absorbs xM in more ways.

Networks of one block. A residual network of one block is a block with the reads EW_{read} and the writes $W_{\text{write}}D$, plus the linear term XED: it computes $X \mapsto XED + f(XEW_{\text{read}})\, W_{\text{write}}D$. The proof of Theorem 3.1(i) holds for such rectangular reads, writes and linear terms.

Proposition 3.2 (Networks of one block)**.** *Let σ be ReLU or GELU. For almost every choice of weights, a residual network of one block equals no skipless network of one block of the same width.*

Proof in Appendix A.1.

3.2 ReLU^2, gated and tanh blocks: no width suffices

A block absorbs xM if a block of the same kind satisfies (2): $xM + B_\theta(x) = B_{\theta'}(x)$ for all x.

Theorem 3.3 (No width suffices)**.** *Let B_θ be (a) a block that is positively homogeneous of degree $k \neq 1$, (b) a gated block whose gate vanishes at 0 and is differentiable there, or (c) a tanh block. ReLU^2 and ReGLU blocks fall under (a), SwiGLU and GeGLU blocks under (b). For every $M \neq 0$, no block $B_{\theta'}$ of the same kind and of any width satisfies (2).*

Proof in Appendix A.1.

The three cases of Figure 1 are three one-line proofs. A homogeneous block satisfies $B_\theta(tu) = t^k B_\theta(u)$ for $t > 0$, so along a ray (2) reads $t\,uM = t^k\big(B_{\theta'}(u) - B_\theta(u)\big)$, which forces $uM = 0$ for every u when $k \neq 1$. A gate that vanishes at 0 makes every unit of order t^2 near the origin, so both blocks have derivative 0 there, while xM has derivative M. A tanh block is bounded, and xM is not.

Degree $k = 1$ is the ReLU case of Theorem 3.1. The condition on the gate is needed: the logistic sigmoid, the gate of the original GLU, does not vanish at 0, so a unit with a zero gate read is linear, and a block absorbs its skip with d_{model} such units.

3.3 Attention blocks: no number of heads suffices

An *attention block* has H heads of dimension d_h. Head i has the matrices $W_{Q,i}, W_{K,i}, W_{V,i} \in \mathbb{R}^{d_{\text{model}} \times d_h}$ and $W_{O,i} \in \mathbb{R}^{d_h \times d_{\text{model}}}$, and the parameters θ of the block are the matrices of all its heads. The block maps $h \in \mathbb{R}^{T \times d_{\text{model}}}$, whose rows are the $T \geq 2$ tokens, to

$$B_\theta(h) = \sum_{i=1}^{H} P_{\Gamma_i}(h)\, h\, W_{V,i} W_{O,i}, \qquad P_\Gamma(h) = \operatorname{softmax}\big(h\, \Gamma h^\top\big), \qquad \Gamma_i = W_{Q,i} W_{K,i}^\top / \sqrt{d_h}. \tag{4}$$

The entries of the $T \times T$ matrix $h\,\Gamma_i h^\top$ are the scores of head i. The softmax acts on each of its rows, with or without a causal mask, so each row of $P_\Gamma(h)$ is a probability vector. The read of the block holds the matrices $W_{Q,i}$, $W_{K,i}$ and $W_{V,i}$, and its write the matrices $W_{O,i}$. Head i has the score matrix Γ_i and the value–output product $W_{V,i} W_{O,i}$.

Two inputs show that an attention block absorbs no linear term. Suppose that $hM + B_\theta(h) = B_{\theta'}(h)$ for all h. If all tokens are equal, the attention weights do not matter, so the sums of the value–output products over the heads of the two blocks must differ by M. Now let one token be zero, placed last when the mask is causal. Its query is zero, so every head gives it the average of the tokens it sees, and the two blocks differ there by this mean times M. The skip adds zero to this token, so this mean times M vanishes for every choice of the other tokens, and $M = 0$.

Theorem 3.4 (No absorption in attention blocks)**.** *Let θ and θ' be parameters of attention blocks with any numbers of heads and any head dimensions, and let $M \neq 0$ be a $d_{\text{model}} \times d_{\text{model}}$ matrix.*
(i) *For almost every $h \in \mathbb{R}^{T \times d_{\text{model}}}$, $hM + B_\theta(h) \neq B_{\theta'}(h)$.*
(ii) *Some h whose rows have norm at most 1 has a row in which $hM + B_\theta(h)$ and $B_{\theta'}(h)$ differ by at least $\|M\|/3$, where $\|M\|$ is the spectral norm.*

A special case of Theorem A.13, proved in Appendix A.2.

The two inputs give one h at which the two sides differ, and both sides are analytic in h, which gives (i). Part (ii) bounds approximate absorption: on inputs with rows of norm at most 1, no attention block comes closer than $\|M\|/3$ to absorbing hM. The proof uses only that each row of attention weights sums to 1 and is uniform for a zero query, so, without biases, the theorem also holds with rotary embeddings. The same two inputs show that a residual network of one attention block equals no skipless network of one attention block, with any number of heads (Theorem A.13(iii)).

4 Networks

The negative answers carry over to networks: without normalization, no residual network of ReLU^2, gated or tanh *blocks equals a skipless network, nor does one of attention blocks on long enough sequences. Under RMSNorm, almost no residual network equals a skipless one, whatever its blocks. ReLU and GELU networks without normalization are the exception: d_{model} more units per block suffice.*

Recall that a network embeds its input as $h_0 = XE$ and reads out $y = h_L D$, and that its skips add $h_{\ell-1}$ at every block.

4.1 ReLU^2, gated and tanh networks: never equal

Theorem 4.1 (No equality for ReLU^2, gated and tanh networks)**.** *Let the residual network have $ED \neq 0$.*
(i) *Without normalization or with pre-RMSNorm and $\varepsilon > 0$, let the blocks be gated blocks whose gate vanishes at 0 and is differentiable there, or blocks that are positively homogeneous of degree $k > 1$.*

No skipless network of blocks of the same kind, of any depth and widths, equals the residual network. **(ii)** *Without normalization, no skipless network of* $\tanh$ *blocks, of any depth and widths, equals a residual network of* $\tanh$ *blocks.*

Proof in Appendix A.1.

The reasons are those of Figure 1, now for the whole network. A gated block, and a homogeneous block of degree $k > 1$, has derivative 0 at the origin. So every skipless network of them has Jacobian 0 at the input $X = 0$, while a residual network has Jacobian $ED \neq 0$ there, or $ED/\sqrt{\varepsilon}$ under RMSNorm. This comparison of derivatives needs $k > 1$. A skipless $\tanh$ network is bounded, while a residual one grows like XED along rays. With $E = D = I$, the theorem covers compositions of blocks.

4.2 Attention networks: never equal on long sequences

The comparison at $X = 0$ also settles networks of attention blocks. Near $h = 0$ the scores of an attention block are of second order, so the block acts as $h \mapsto UhV$: the $T \times T$ matrix U replaces each token by the average of the tokens it attends to, and V is the sum of the value–output products of the heads. At $X = 0$, a skipless network of L' attention blocks therefore has the Jacobian $\delta \mapsto U^{L'} \delta \, \Lambda'$ for some matrix Λ': an output token sees the input only through L' rounds of averaging. A residual network keeps the term $\delta \mapsto \delta \, ED$, with no averaging.

Theorem 4.2 (Attention networks on long and short sequences)**.** *Without normalization or with pre-RMSNorm and $\varepsilon > 0$, let a residual network of L attention blocks have $ED \neq 0$. Consider skipless networks of $L' \geq 1$ attention blocks, with any numbers of heads and any head dimensions.*
(i) *If the attention has no mask, no such skipless network equals the residual one. With a causal mask on T tokens, the same holds when $T \geq L + 2$, and when $T = L + 1$ and $L' \leq L$.*
(ii) *For almost every choice of weights of the residual network, no such skipless network equals it, whatever L'.*
(iii) Short sequences. *Without normalization and with a causal mask, in every other case, $T \leq L$ or $T = L + 1 \leq L'$, some residual network of L blocks with $ED \neq 0$ equals a skipless network of L' blocks.*

Proof in Appendix A.2.

In the cases of (i), no number of heads helps. Neither does depth, once a sequence has at least two more tokens than the residual network has blocks, as in language models. Part (iii) shows that the conditions of (i) cannot be weakened. Under a causal mask, token t averages the first t tokens, so U has the T eigenvalues $1, 1/2, \ldots, 1/T$. At $X = 0$ the Jacobian of a residual network of L blocks is a polynomial in U of degree at most L with a nonzero constant term, and it cannot agree with a multiple of $U^{L'}$ at $L + 2$ or more eigenvalues. At fewer it can, in the cases of (iii). For $T = 2$ and heads whose scores vanish, the residual block $h \mapsto h - 3Uh$ equals the composition $h \mapsto -2U^2 h$ of two skipless blocks. Under pre-RMSNorm, equality in the cases of (iii) fails for almost all weights, by (ii); whether it ever holds is open.

4.3 Networks under RMSNorm: almost never equal

Under RMSNorm the answer no longer depends on the kind of block. A skip keeps the norm of each token, and a skipless network loses it at its first normalization.

Theorem 4.3 (Almost no equality under RMSNorm)**.** *Consider pre-RMSNorm networks, with $\varepsilon \geq 0$ and blocks of any kind and width. For almost every choice of weights, a residual network equals no skipless network, whatever its depth, widths and kinds of blocks.*

Proof in Appendix A.3.

The proof compares rates of convergence along rays tX, as $t \to \infty$. A skipless network with embedding E' depends on t only through $N(tXE')$, that is, through ε/t^2, so its output converges at rate $1/t^2$ or faster. In a residual network the skips carry tXE to the last normalization, and every block reads, up to $O(1/t)$, the same normalized input and adds a bounded output. These outputs turn

each token by an angle of order $1/t$, so the output of the network converges only at rate $1/t$, unless in every token they add up to a multiple of the token itself. For almost every choice of weights, at some input they do not. The theorem covers pre-RMSNorm transformers, which alternate attention and MLP blocks.

4.4 ReLU and GELU networks without normalization: d_{model} more units per block suffice

Without normalization, ReLU and GELU networks are the one case in which almost every residual network equals a skipless one: the construction of Section 3.1 works block by block.

Proposition 4.4 (d_{model} more units per block suffice)**.** *Let σ be ReLU or GELU. Without normalization, a residual network whose reads all have rank d_{model} equals a skipless network with d_{model} more hidden units in every block.*

Proof in Appendix A.1.

Some of these units are needed, at least in the first block.

Theorem 4.5 (The first block needs $2d_{\text{model}}$ units)**.** *Consider networks of ReLU blocks, or of GELU blocks, without normalization. For almost every choice of weights, a residual network with hidden dimension d_{model} equals no skipless network whose first block has fewer than $2d_{\text{model}}$ hidden units, whatever its depth and its other widths.*

Proof in Appendix A.1.

For ReLU the proof compares ranks. Near almost every input, a generic residual ReLU network is a linear map of rank d_{model}. No unit of the first block is active at both an input and its negative, so with fewer than $2d_{\text{model}}$ units a skipless network has fewer than d_{model} active units at one of the two, and some direction of inputs leaves it unchanged there. GELU reduces to ReLU along rays: $|\mathrm{GELU}(tz) - t\,\mathrm{ReLU}(z)| < 0.17$ for all z and all $t > 0$, so as $t \to \infty$ the output of a GELU network at the input tX, divided by t, tends to the output at X of the ReLU network with the same weights. Two GELU networks that are equal therefore stay equal with ReLU in place of GELU.

For the first block, the two bounds meet when the residual first block has d_{model} units. When the residual blocks have $2d_{\text{model}}$ units or more, as in the usual width $4d_{\text{model}}$, the theorem allows a skipless network of the same widths. The rest of this section bounds the extra units in two cases, and what remains is open.

Injective residual blocks: every skipless block needs $2d_{\text{model}}$ units.

Proposition 4.6 (Injective blocks: $2d_{\text{model}}$ units in every block)**.** *Let σ be ReLU or GELU, without normalization, and let E and D have rank d_{model}. Suppose that every block of the residual network is injective on tokens when σ is replaced by ReLU: $x + B_\ell(x) \neq x' + B_\ell(x')$ for all tokens $x \neq x'$. Then every skipless network of σ blocks, of any depth and with the same d_{model}, that equals the residual network has at least $2d_{\text{model}}$ hidden units in every block.*

Proof in Appendix A.1.

The condition holds when every block has $\|W_{\text{read}}\|\,\|W_{\text{write}}\| < 1$, in spectral norm: the ReLU block B is then a contraction, so $x \mapsto x + B(x)$ is injective. This is an open condition on the weights, and it holds for every network whose writes are small enough for fixed reads, E and D. When every block of the residual network also has d_{model} hidden units and an invertible read, an equal skipless network of the same depth needs exactly d_{model} extra units in every block, by Proposition 4.4.

The proof turns on injectivity. If every residual block is injective, the residual network maps distinct embedded tokens xE to distinct outputs. So every block of an equal skipless network must map distinct tokens to distinct tokens, and a ReLU block without biases does so only with $2d_{\text{model}}$ units, since no unit is active at both x and $-x$.

For ReLU networks the condition cannot be dropped. On another open set of weights, around the first block $x \mapsto x + 2\,\mathrm{ReLU}(-x)$, which replaces every coordinate of x by its absolute value, the first block is not injective, the units of later blocks are never active, and d_{model} extra units in the first

block suffice at every depth (Remark A.12). For GELU networks without the condition, the question is open.

Wide residual blocks: a bound from counting dimensions. When the residual blocks have $2d_{\text{model}}$ units or more, Proposition 4.6 says nothing. Counting dimensions still bounds the extra units of ReLU networks. Let the residual network have P weights and s hidden units. Its function does not change under one change of basis of the hidden states, nor when the read of a unit is multiplied by $c > 0$ and its write by $1/c$. So on any finite set of inputs, the Jacobian of the outputs with respect to the weights has rank at most $P - d_{\text{model}}^2 - s$ (Karbevski and Mijoski, 2026). We call a residual ReLU network *full* if this rank is reached on some finite set of inputs at which no unit has argument 0: its only redundancy is then these symmetries.

Proposition 4.7 (A lower bound from counting dimensions)**.** *Consider ReLU networks of L blocks without normalization. The full residual networks form an open set of weights. For almost every full residual network, every skipless network of L blocks with the same d_{model} that equals it has, in total, at least $Ld_{\text{model}}^2/(2d_{\text{model}} - 1)$ more hidden units.*

Proof in Appendix A.1.

A skipless network has a change of basis at each of its $L + 1$ hidden states, so it must make up Ld_{model}^2 dimensions, and an extra ReLU unit adds $2d_{\text{model}} - 1$. The bound is about half of the Ld_{model} extra units of Proposition 4.4. Full networks exist: every network with generic weights is full once all its writes are scaled by a small enough factor. Not every network is full, since on another open set of weights a unit is never active (Karbevski and Mijoski, 2026). For GELU networks the same count gives at least $Ld_{\text{model}}/2$ extra units for almost every residual network (Karbevski and Mijoski, 2026).

5 Skipless networks are limits of residual ones

Without normalization, every skipless network is a limit of residual networks of the same widths.

Proposition 5.1 (Skipless networks as limits of residual networks)**.** *Let F be a skipless network whose blocks are locally Lipschitz.*
(i) Without normalization. *The network F is a limit of residual networks with blocks of the same kinds and widths, uniformly on bounded sets of inputs.*
(ii) Under pre-RMSNorm with $\varepsilon = 0$. *The network F is a limit of residual networks with blocks of the same kinds and widths, uniformly on compact sets of inputs at which F is defined.*

Proof in Appendix A.4.

Idea of the proof. The proof scales the reads of block ℓ by $t^{1-\ell}$, its writes by t^ℓ and the readout by t^{-L}. Block ℓ then sees the hidden state of F, up to $O(1/t)$, and outweighs its skip by a factor t, so the residual network computes $F + O(1/t)$ (Remark A.16). The large constants sit in the writes, so no activation sees a large argument, and the construction applies alike to $\tanh$, GELU, ReLU, ReLU^2, gated and attention blocks. In this construction the weights diverge as $t \to \infty$. So without normalization, residual networks come arbitrarily close to every skipless network on bounded sets of inputs, even where none equals it.

Under RMSNorm. With $\varepsilon = 0$, RMSNorm ignores the size of the hidden states, so only the writes grow. With $\varepsilon > 0$ the construction fails, since on a hidden state of size t^ℓ RMSNorm acts as RMSNorm with $\varepsilon/t^{2\ell}$; whether skipless networks are limits of residual ones there is open.

6 Related work

Expressivity of residual networks. Lin and Jegelka (2018) show that residual ReLU networks with one hidden unit per block are universal approximators, which skipless ReLU networks with at most d_{in} units per layer are not (Appendix B); in their count, the identity map of a skip takes d_{model}

hidden units. For blocks without biases and with at least d_{model} units, Theorem 3.1(ii) shows that this count of d_{model} units is exact for almost every ReLU or GELU block, and Theorem 3.3 that no count works for ReLU^2, gated and $\tanh$ blocks. With biases, as in their blocks, the count stays exact for almost every ReLU or GELU block with at least d_{model} units (Theorem C.2 in Appendix C). Hardt and Ma (2017) show that deep linear residual networks have no spurious local optima, and that residual ReLU networks can fit any finite sample once they have more parameters than samples. These results concern approximation, optimization or finite samples; ours concern exact equality of functions, block by block and activation by activation.

Identifiability. For $\tanh$ networks, the function determines the weights up to sign flips and permutations of hidden units (Fefferman, 1994; Sussmann, 1992), and similar statements hold for other activations (Vlačić and Bölcskei, 2022) and for ReLU networks (Phuong and Lampert, 2020; Rolnick and Kording, 2020). For two-layer ReLU networks with one output, Petzka et al. (2020) add the sign flips that change a linear term, the mechanism of Theorem 3.1(i); pairs of ReLU units with opposite input weights also appear in the lower layers of trained convolutional networks (Shang et al., 2016). Our proofs identify the units of a block from its function in the same way, for several outputs and for GELU, and then determine what is left for the linear term of the skip. For attention, the function of a block with three or more tokens determines the score matrices and the value–output products of its heads, up to merging heads that share a score matrix and dropping heads that output zero (Proposition A.15); Tran et al. (2026) determine in the same way when two multi-head attention layers are equal, also with rotary embeddings.

Training without skips. Skipless networks train with weights parametrized as the identity plus a learned matrix (Zagoruyko and Komodakis, 2017), or with careful initialization or activation shaping (Martens et al., 2021; Zhang et al., 2022), and skipless transformers with modified attention (He et al., 2023; Noci et al., 2023) or simplified blocks (He and Hofmann, 2024). Ji et al. (2026) train a skipless ViT-B that outperforms the residual one. Our results say when a skipless block or network can compute exactly what a residual one computes, and with how many units.

7 Conclusion

Without normalization or biases, a trained residual ReLU or GELU network whose reads have rank d_{model} can be rewritten exactly as a skipless network with d_{model} more units per block. A residual ReLU^2, gated or $\tanh$ network with $ED \neq 0$ cannot be rewritten at any width, nor can a residual attention network on long enough sequences, and under pre-RMSNorm almost no residual network can, whatever its blocks; this includes pre-RMSNorm transformers. With biases and without normalization, residual ReLU^2 and gated networks can be rewritten too (Remark C.5). Yet without normalization every skipless network is a limit of residual networks of the same widths. All of this concerns exact equality at every input; Table 1 lists the cases.

Open questions. The unknown cells and the conditions of Table 1 leave three questions, and a fourth lies beyond the table.

(1) Networks of attention blocks under pre-RMSNorm with $\varepsilon > 0$ and a causal mask, with at most one more token than blocks. Without normalization some of them are equal (Theorem 4.2(iii)); under RMSNorm equality fails for almost all weights, and whether it ever holds is open.

(2) How many extra units ReLU and GELU networks need, without normalization, when the blocks are wider than d_{model}, have reads of rank below d_{model}, or are not injective. For almost every full ReLU network, counting dimensions gives about half of the Ld_{model} units that suffice (Proposition 4.7); whether Ld_{model} are needed when every block is injective and the blocks are wider than d_{model} is open.

(3) Skipless networks as limits of residual networks under RMSNorm with $\varepsilon > 0$.

(4) Approximation: how many units a skip is worth at a given error on bounded inputs, and whether residual networks are limits of skipless ones.

Use of AI

The conception of the study is the authors' own. One author obtained their results, proofs and experiments working with AI assistants, through continuous conversation; the assistants were used to find, verify and simplify proofs and to write the code of the experiments, which the authors validated.

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A Proofs

A.1 MLP blocks and networks (Sections 3 and 4)

This subsection proves the results of Sections 3 and 4 on MLP blocks and networks, for the question (2). The auxiliary results come first; each result of the main text is then recalled right before its proof. Below we write B for B_θ, and B' for $B_{\theta'}$ with $\theta' = (W'_{\text{read}}, W'_{\text{write}})$.

Overview. Theorem 3.3 compares orders. Along rays from the origin a linear skip scales like λ and a block that is homogeneous of degree $k \neq 1$ like λ^k, and near the origin a gated block is of second order, so the two sides of the absorption equation cannot agree (Propositions A.1 and A.2); comparing Jacobians at the origin extends this to networks of any depth, Theorem 4.1 (Corollary A.3). Theorem 3.1(i) is the substantial case. The absorption equation splits into its odd and even parts in x. The even part determines the hidden units of the absorbing block up to sign, and for ReLU up to a positive rescaling, through the kinks for ReLU (Theorem A.4) and through linear independence of ridge functions for GELU (Lemma A.5 and Theorem A.7). The odd part of ReLU and GELU is exactly $z/2$, so the odd part of the equation is linear and reads off the set S of units whose signs flip; conversely, flipping the signs of the units in S absorbs the skip whenever the condition holds. Corollary A.9 extends the criterion to every linear term xM, of any shape, and the generic clause holds because the condition is a finite union of algebraic sets of positive codimension (Corollary A.10). Theorem 3.1(ii) counts the extra units that the matching requires (Proposition A.11). Proposition 4.4 applies Theorem 3.1(ii) block by block. Theorem 4.5 compares the ranks of the linear pieces of a ReLU network at an input and at its negative, and reduces GELU to ReLU along rays. Proposition 4.6 passes injectivity from the residual network to every block of an equal skipless network, and Proposition 4.7 compares the ranks of the Jacobians of the two networks with respect to their weights.

The arguments show that the function of a block determines its parameters, in the manner used for $\tanh$ networks by Sussmann (1992) and Fefferman (1994) and for ReLU networks by Phuong and Lampert (2020), Petzka et al. (2020) and Rolnick and Kording (2020): kink alignment for ReLU, ridge-function independence for GELU. For two-layer ReLU networks with a single output, Petzka et al. (2020, Section 3) already give the corresponding sign flips, show that irreducible networks computing the same function differ only by these flips, permutations and positive rescalings (their Theorem 1), and state the matching criterion for removing a linear term.

Setting. Throughout, $W_{\text{read}} \in \mathbb{R}^{d_{\text{model}} \times m}$ and $W_{\text{write}} \in \mathbb{R}^{m \times d_{\text{model}}}$, σ acts coordinatewise, w_i is column i of W_{read} and $\hat{w}_i$ is row i of W_{write}, and inputs x are row vectors. For $S \subseteq \{1, \dots, m\}$, $W_{\text{read}}[:, S]$ is the submatrix of columns in S, $W_{\text{write}}[S, :]$ the submatrix of rows in S, and Π_S the diagonal projector with $(\Pi_S)_{ii} = 1$ exactly when $i \in S$, so that $W_{\text{read}} \Pi_S W_{\text{write}} =$

$W_{\text{read}}[:, S]W_{\text{write}}[S, :]$. A family of vectors is *pairwise non-collinear* if no one of them is a scalar multiple of another, and a block is *nondegenerate* if the columns of W_{read} are nonzero and pairwise non-collinear and the rows of W_{write} are nonzero. The question is whether there is a block $B'(x) = \sigma(xW'_{\text{read}})W'_{\text{write}}$ of the same width, with $W'_{\text{read}} \in \mathbb{R}^{d_{\text{model}} \times m}$ and $W'_{\text{write}} \in \mathbb{R}^{m \times d_{\text{model}}}$, such that

$$x + \sigma(xW_{\text{read}})W_{\text{write}} = \sigma(xW'_{\text{read}})W'_{\text{write}} \qquad \text{for all } x \in \mathbb{R}^{1 \times d_{\text{model}}}. \tag{5}$$

Proposition A.1 (Homogeneous blocks)**.** *Let B and B' be blocks that are positively homogeneous of degree $k \neq 1$, of any widths, and let $M \neq 0$. Then $xM + B(x) = B'(x)$ fails for some x.*

Examples are ReLU^2 and ReGLU blocks without biases, both of degree 2.

Proof. Outline. We compare the two sides along rays through the origin: a term of degree one and a term of degree $k \neq 1$ cannot agree at every scale.

Suppose it holds for all x and evaluate at λx with $\lambda > 0$: $\lambda xM + \lambda^k B(x) = \lambda^k B'(x)$. Dividing by λ gives $xM = \lambda^{k-1}\big(B'(x) - B(x)\big)$, and $\lambda^{k-1} \to 0$ as $\lambda \to 0$ if $k > 1$ and as $\lambda \to \infty$ if $k < 1$. So $xM = 0$ for all x, which contradicts $M \neq 0$. □

Proposition A.2 (Gated blocks)**.** *Let B and B' be gated blocks without biases, of any widths, whose gates are differentiable at 0 and vanish there, and let $M \neq 0$. Then $xM + B(x) = B'(x)$ fails for some x.*

Examples are SwiGLU and GeGLU blocks, which Shazeer (2020) defines without biases.

Proof. Outline. Near the origin a gated block is of second order in x, while xM is of first order.

A unit of a gated block computes $g(u \cdot x)\,(v \cdot x)$. Since $g(0) = 0$ and g is differentiable at 0, $|g(u \cdot x)| \leq c\,\|x\|$ for small $\|x\|$, so the unit is $O(\|x\|^2)$, and so are B and B'. If $xM + B(x) = B'(x)$ for all x, then $xM = O(\|x\|^2)$ as $x \to 0$. Along a ray $x = tz$ this reads $t\,zM = O(t^2)$, so $zM = 0$ for every z, which contradicts $M \neq 0$. □

Corollary A.3 (Any depth)**.** *Let every block be as in Proposition A.1 with $k > 1$ or as in Proposition A.2, of any widths. Let F be a composition of residual blocks $h \mapsto h + B_\ell(h)$ and H a composition of one or more skipless blocks. Then $F \neq H$. With embeddings and readouts, the residual network $X \mapsto F(XE)D$ differs from the skipless network $X \mapsto H(XE')D'$ whenever $ED \neq 0$, also with pre-RMSNorm and $\varepsilon > 0$. Without normalization the same holds for* tanh *blocks.*

Proof. Each such block maps 0 to 0 and has Jacobian 0 there, because it is $O(\|x\|^k)$ with $k > 1$, respectively $O(\|x\|^2)$. By the chain rule F has Jacobian I at the origin and H has Jacobian 0; with embeddings and readouts these become ED and 0. Under pre-RMSNorm with $\varepsilon > 0$, $N(0) = 0$ and N has Jacobian $I/\sqrt{\varepsilon}$ at 0, so every block, read through N, still has Jacobian 0 there, and the residual and the skipless network have Jacobians $ED/\sqrt{\varepsilon}$ and 0 at the origin. For tanh blocks without normalization, H and $F(h) - h$ are bounded, so $F(XE)D - H(XE')D'$ differs from XED by a bounded function and does not vanish when $ED \neq 0$. □

Parity. The remaining activations have an odd part that is exactly linear. For ReLU, $\mathrm{ReLU}(z) = |z|/2 + z/2$. For GELU, $\mathrm{GELU}(z) = z\Phi(z)$ with Φ the standard normal distribution function, and since $\Phi(-z) = 1 - \Phi(z)$,

$$\mathrm{GELU}(z) = \Psi(z) + z/2, \qquad \Psi(z) := z\big(\Phi(z) - \tfrac{1}{2}\big), \qquad \Psi(-z) = \Psi(z).$$

The even parts, $|z|/2$ and Ψ, match the hidden units of the two blocks; the odd part, $z/2$, turns a sign flip into a linear term and gives the absorption condition.

Theorem A.4 (Identity absorption for ReLU)**.** *Let $\sigma = \mathrm{ReLU}$, $d_{\text{model}} \geq 2$, and let the block be nondegenerate. Then (5) has a solution if and only if*

$$W_{\text{read}}[:, S]\, W_{\text{write}}[S, :] = -I_{d_{\text{model}}} \quad \textit{for some } S. \tag{6}$$

Proof. Outline. Sufficiency is a construction: flipping the signs of the hidden units indexed by S produces the missing identity term. For necessity we split (5) into its odd and even parts in x, use the kinks of the even part to match the hidden units of the two blocks one to one, and read the index set S off the signs of the matching.

Sufficiency. Let $F = I_m - 2\Pi_S$, a diagonal sign matrix, and set $W'_{\text{read}} = W_{\text{read}}F$, $W'_{\text{write}} = W_{\text{write}}$. Since $|xW_{\text{read}}F| = |xW_{\text{read}}|$ coordinatewise,

$$\begin{aligned}\text{ReLU}(xW'_{\text{read}})W'_{\text{write}} &= \tfrac{1}{2}\big(xW_{\text{read}}F + |xW_{\text{read}}|\big)W_{\text{write}},\\ x + \text{ReLU}(xW_{\text{read}})W_{\text{write}} &= \tfrac{1}{2}\big(xW_{\text{read}} + |xW_{\text{read}}|\big)W_{\text{write}} + x,\end{aligned}$$

and the two agree if and only if $\frac{1}{2}xW_{\text{read}}(F - I_m)W_{\text{write}} = x$ for all x, that is $-W_{\text{read}}\Pi_S W_{\text{write}} = I_{d_{\text{model}}}$, which is (6).

Necessity. Suppose (5) holds and split both sides into odd and even parts in x:

$$W'_{\text{read}}W'_{\text{write}} = W_{\text{read}}W_{\text{write}} + 2I_{d_{\text{model}}}, \qquad |xW'_{\text{read}}|\,W'_{\text{write}} = |xW_{\text{read}}|\,W_{\text{write}} \ \text{ for all } x. \tag{7}$$

Step 1: alignment of kinks. The map $x \mapsto |xW_{\text{read}}|\,W_{\text{write}}$ fails to be differentiable exactly on the union of the hyperplanes $\{xw_i = 0\}$; these are m distinct hyperplanes, each carrying a genuine kink since $\hat{w}_i \neq 0$. By the second identity in (7) the map $x \mapsto |xW'_{\text{read}}|\,W'_{\text{write}}$ has the same non-differentiability locus. Each of the m kink hyperplanes $\{x : xw_i = 0\}$ of the target needs at least one unit of B' whose kink lies on it, and B' has only m units; so each hyperplane gets exactly one unit, no two units share a hyperplane, and no kinks cancel. After relabeling, $W'_{\text{read}} = W_{\text{read}}C$ with $C = \text{diag}(c_1, \ldots, c_m)$, all $c_i \neq 0$. *Step 2: coefficients.* Substituting, $|xW_{\text{read}}C| = |xW_{\text{read}}|\,|C|$, so $|xW_{\text{read}}|\,|C|\,W'_{\text{write}} = |xW_{\text{read}}|\,W_{\text{write}}$ for all x. For $d_{\text{model}} \geq 2$ and pairwise non-collinear columns, the functions $|xw_i|$ are linearly independent modulo linear functions, because their kink hyperplanes are distinct; hence $|C|\,W'_{\text{write}} = W_{\text{write}}$ and $W'_{\text{write}} = |C|^{-1}W_{\text{write}}$. *Step 3: signs.* The first identity in (7) becomes $W_{\text{read}}C|C|^{-1}W_{\text{write}} = W_{\text{read}}W_{\text{write}} + 2I_{d_{\text{model}}}$. With $\Sigma = C|C|^{-1} = \text{diag}(\text{sign}\, c_i)$ this reads $W_{\text{read}}(\Sigma - I_m)W_{\text{write}} = 2I_{d_{\text{model}}}$; the diagonal of $\Sigma - I_m$ has entries 0 or -2, so with $S = \{i : c_i < 0\}$ we get $W_{\text{read}}\Pi_S W_{\text{write}} = -I_{d_{\text{model}}}$, and $|S| \geq d_{\text{model}}$ by the rank of $I_{d_{\text{model}}}$. □

The GELU case needs one classical fact about ridge functions, whose short proof we include.

Lemma A.5 (Vanishing sums of ridge functions)**.** *Let $a_1, \ldots, a_r \in \mathbb{R}^{d_{\text{model}}}$ be nonzero and pairwise non-collinear, and let $g_1, \ldots, g_r : \mathbb{R} \to \mathbb{R}^q$ be $r-1$ times continuously differentiable with $\sum_i g_i(a_i \cdot x) = 0$ for all $x \in \mathbb{R}^{d_{\text{model}}}$. Then every g_i is a polynomial of degree at most $r-2$.*

Proof. Fix i. For each $j \neq i$ choose $u_j \in \mathbb{R}^{d_{\text{model}}}$ with $u_j \cdot a_j = 0$ and $u_j \cdot a_i \neq 0$, which exists because a_i is not a multiple of a_j. The directional derivative ∂_{u_j} annihilates every function of $a_j \cdot x$ and maps a function of $a_k \cdot x$ to a function of $a_k \cdot x$. Applying the product of the commuting operators ∂_{u_j}, $j \neq i$, to the sum therefore leaves $\prod_{j \neq i}(u_j \cdot a_i)\, g_i^{(r-1)}(a_i \cdot x) = 0$. Since $a_i \neq 0$, $a_i \cdot x$ takes every real value, so $g_i^{(r-1)} = 0$. □

Lemma A.6 (Distinct scales)**.** *Let $\psi : \mathbb{R} \to \mathbb{R}$ be real-analytic and not a polynomial. Let $\alpha_1, \ldots, \alpha_n > 0$ be distinct and $a_1, \ldots, a_n \in \mathbb{R}^q$. If $\sum_k a_k\, \psi(\alpha_k t)$ is a polynomial in t, then every a_k is zero.*

Proof. Near $t = 0$, $\psi(t) = \sum_j \psi_j t^j$. Infinitely many ψ_j are nonzero, as otherwise ψ would be a polynomial near 0, hence everywhere. The coefficient of t^j in the sum is $\psi_j \sum_k a_k \alpha_k^j$, and it vanishes for all large j. So $\sum_k a_k \alpha_k^j = 0$ for n distinct exponents j, and a generalized Vandermonde matrix with distinct positive nodes is invertible. □

Theorem A.7 (Identity absorption for GELU)**.** *Let $\sigma = \text{GELU}$ and let the block be nondegenerate. Then (5) has a solution if and only if (6) holds.*

Proof. Outline. The proof follows that of Theorem A.4. Linear independence of ridge functions replaces the kink argument, and one extra step shows that matched hidden units have the same scale.

Sufficiency. With $F = I_m - 2\Pi_S$, $W'_{\text{read}} = W_{\text{read}}F$ and $W'_{\text{write}} = W_{\text{write}}$, evenness of Ψ gives $\Psi(xW_{\text{read}}F) = \Psi(xW_{\text{read}})$, so

$$\begin{aligned}\text{GELU}(xW'_{\text{read}})W'_{\text{write}} &= \Psi(xW_{\text{read}})W_{\text{write}} + \tfrac{1}{2}xW_{\text{read}}FW_{\text{write}} \\ &= \text{GELU}(xW_{\text{read}})W_{\text{write}} - xW_{\text{read}}\Pi_S W_{\text{write}} \\ &= \text{GELU}(xW_{\text{read}})W_{\text{write}} + x.\end{aligned}$$

Necessity. Splitting (5) into even and odd parts gives

$$\Psi(xW'_{\text{read}})W'_{\text{write}} = \Psi(xW_{\text{read}})W_{\text{write}} \text{ for all } x, \qquad W'_{\text{read}}W'_{\text{write}} = W_{\text{read}}W_{\text{write}} + 2I_{d_{\text{model}}}. \tag{8}$$

Step 1: alignment of directions. Ψ is analytic and not a polynomial. Zero columns of W'_{read} contribute $\Psi(0) = 0$; group the other columns of W'_{read} and the columns of W_{read} into classes of collinear vectors. In the first identity of (8) the terms of each class form one ridge function in the direction of the class, and these ridge functions sum to zero, so each is a polynomial by Lemma A.5. The class of a column w_i of W_{read} contains no other column of W_{read}, so if it contained no column of W'_{read} its ridge function would be $-\Psi(xw_i)\,\hat{w}_i$, which is not a polynomial because $\hat{w}_i \neq 0$. Hence each of the m classes of the columns of W_{read} contains a nonzero column of W'_{read}, and since W'_{read} has only m columns, after relabeling $W'_{\text{read}} = W_{\text{read}}C$ with $C = \text{diag}(c_i)$, $c_i \neq 0$. *Step 2: unit scale.* Collecting the terms of direction w_i on both sides gives, for each i, the ridge function $g_i(xw_i)$ with

$$g_i(t) = \Psi(|c_i|\,t)\,(W'_{\text{write}})_{i,:} - \Psi(t)\,(W_{\text{write}})_{i,:},$$

and $\sum_i g_i(xw_i) = 0$. By Lemma A.5 each g_i is a polynomial. If $|c_i| \neq 1$, Lemma A.6 with the scales $|c_i|$ and 1 gives $(W_{\text{write}})_{i,:} = 0$, which nondegeneracy excludes. So $|c_i| = 1$ and $g_i(t) = \Psi(t)\big((W'_{\text{write}})_{i,:} - (W_{\text{write}})_{i,:}\big)$, and Lemma A.6 gives $(W'_{\text{write}})_{i,:} = (W_{\text{write}})_{i,:}$. Therefore $|C| = I_m$ and $W'_{\text{write}} = W_{\text{write}}$. *Step 3: signs.* Identical to the ReLU case: the second identity in (8) becomes $W_{\text{read}}(C - I_m)W_{\text{write}} = 2I_{d_{\text{model}}}$ with $C = \text{diag}(\pm 1)$, so $S = \{i : c_i = -1\}$ satisfies (6) and $|S| \geq d_{\text{model}}$. □

Remark A.8 (The even part). Any activation whose even part ψ is analytic and not a polynomial matches the hidden units in the same way: Steps 1 and 2 of the proof of Theorem A.7 apply with ψ in place of Ψ; zero columns of W'_{read} now contribute a constant, which is a polynomial. Linear independence of the ridge functions alone does not suffice: for $\sigma(z) = z/2 + |z|^3$ the even part is homogeneous, so a unit can be rescaled together with its output row, and the unit scale is not forced. The odd part o of σ enters only through the flips: with the units matched, both directions reduce to $x = -2\sum_{i\in S} o(xw_i)\,\hat{w}_i$, with S the flipped units. For $o(z) = z/2$, that is $\sigma(z) - \sigma(-z) = z$, as for SiLU, whose even part is $\frac{z}{2}\tanh\frac{z}{2}$, this is (6), and $o(z) = \alpha z$ with $\alpha \neq 0$ only rescales it; a polynomial o of higher degree adds the condition that the units in S cancel its terms of degree three and higher, and a smooth o that is not a polynomial admits no S (Lemma A.5), so the skip is never absorbed at the same width.

Corollary A.9 (Linear terms of any shape)**.** *Let σ be ReLU or GELU. Let $p \geq 2$, let $W_{\text{read}} \in \mathbb{R}^{p\times m}$ have nonzero, pairwise non-collinear columns, let $W_{\text{write}} \in \mathbb{R}^{m\times q}$ have nonzero rows, and let $M \in \mathbb{R}^{p\times q}$. Some $W'_{\text{read}} \in \mathbb{R}^{p\times m}$ and $W'_{\text{write}} \in \mathbb{R}^{m\times q}$ satisfy*

$$xM + \sigma(xW_{\text{read}})W_{\text{write}} = \sigma(xW'_{\text{read}})W'_{\text{write}} \qquad \textit{for all } x \in \mathbb{R}^{1\times p}$$

if and only if $W_{\text{read}}[:, S]\,W_{\text{write}}[S, :] = -M$ for some S. For fixed $M \neq 0$, this holds only on a set of weights $(W_{\text{read}}, W_{\text{write}})$ of measure zero.

Proof. The proofs of Theorems A.4 and A.7 apply with M in place of $I_{d_{\text{model}}}$ and rows $x \in \mathbb{R}^{1\times p}$; nothing in them uses that the linear term is the identity. Sufficiency is the same computation, which now ends with $-W_{\text{read}}\Pi_S W_{\text{write}} = M$. For necessity, the even part of the equation does

not contain M, and Steps 1 and 2 use only that the columns of W_{read} are nonzero, pairwise non-collinear vectors of $\mathbb{R}^p$ with $p \geq 2$, and that the rows of W_{write} are nonzero. The odd part reads $W'_{\text{read}} W'_{\text{write}} = W_{\text{read}} W_{\text{write}} + 2M$, and Step 3 turns it into $W_{\text{read}} \Pi_S W_{\text{write}} = -M$. For each of the finitely many S, this is a system of polynomial equations in the weights that fails at $W_{\text{write}} = 0$ when $M \neq 0$, so it holds only on a set of measure zero. □

With $p = q = d_{\text{model}}$ the corollary covers singular M, and with $p = d_{\text{in}}$ and $q = d_{\text{out}}$ networks of one block.

Corollary A.10 (Generic disjointness for one block)**.** *Let σ be ReLU or GELU and $d_{\text{model}} \geq 2$. For generic weights, (5) has no solution, and a skipless block equals no residual block of the same width.*

Proof. Outline. The condition of Theorems A.4 and A.7 is a finite union of algebraic sets of positive codimension, and such a union has measure zero.

For each of the finitely many S, (6) is d_{model}^2 polynomial equations in the weights. They fail at $W_{\text{write}} = 0$, so they define a set of measure zero; the hypotheses of Theorems A.4 and A.7 hold outside another set of measure zero. So a generic residual block is not a skipless block of the same width. Conversely, $\sigma(xW'_{\text{read}})W'_{\text{write}} = x + \sigma(xW_{\text{read}})W_{\text{write}}$ says that the block with weights $(W'_{\text{read}}, W'_{\text{write}})$ absorbs the linear term $-x$. By Corollary A.9 with $M = -I$, a nondegenerate skipless block is therefore a residual block of the same width only if $W'_{\text{read}}[:, S]\, W'_{\text{write}}[S, :] = I$ for some S, again on a set of measure zero. Neither class contains the other. □

Proposition A.11 (d_{model} extra units)**.** *Let σ be ReLU or GELU, let $d_{\text{model}} \geq 2$, and let $M \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$ be invertible.*
(i) *If W_{read} has rank d_{model}, a block B' of width $m + d_{\text{model}}$ satisfies $xM + B(x) = B'(x)$ for all x.*
(ii) *For generic weights no block of width less than $m + d_{\text{model}}$ does.*

Proof. Outline. Two units with opposite reads produce a linear term of rank one, so d_{model} extra units produce xM. Conversely, the matching of the units in Theorems A.4 and A.7 shows that every extra unit adds at most one such term.

(i) As $\sigma(z) - \sigma(-z) = z$, a column w and rows a, b give $\sigma(xw)(a + b) + \sigma(-xw)(-b) = \sigma(xw)\, a + (xw)\, b$. Choose d_{model} units whose columns $w_{i_1}, \ldots, w_{i_{d_{\text{model}}}}$ form an invertible matrix W_K, and let $b_1, \ldots, b_{d_{\text{model}}}$ be the rows of $W_K^{-1} M$. Adding b_k to the row $\hat{w}_{i_k}$, and a unit with column $-w_{i_k}$ and row $-b_k$, for each k, gives a block of width $m + d_{\text{model}}$ that computes $\sigma(xW_{\text{read}})W_{\text{write}} + xW_K W_K^{-1} M$.

(ii) Outside a set of measure zero the block is nondegenerate, and for each of the finitely many sets S of units the determinant of $M + W_{\text{read}}[:, S] W_{\text{write}}[S, :]$ is nonzero, since it is a polynomial in the weights equal to $\det M \neq 0$ at $W_{\text{write}} = 0$. Let a block B' of width m', with columns w'_j and rows $\hat{w}'_j$, satisfy $xM + B(x) = B'(x)$. Its odd part gives $W'_{\text{read}} W'_{\text{write}} = W_{\text{read}} W_{\text{write}} + 2M$, and its even part $|xW'_{\text{read}}|\, W'_{\text{write}} = |xW_{\text{read}}|\, W_{\text{write}}$ for ReLU and $\Psi(xW'_{\text{read}})W'_{\text{write}} = \Psi(xW_{\text{read}})W_{\text{write}}$ for GELU. Group the nonzero columns w'_j and the columns w_i into classes of collinear vectors, write $w'_j = s_j u$ with a unit column u of the class, $u = w_i / \|w_i\|$ in the class of w_i, and let the row P_u be $\|w_i\| \hat{w}_i$ there and 0 in a class without a column of W_{read}. Below, sums over j run over the columns w'_j of the class of u. The even part then forces, class by class, $\sum_j |s_j|\, \hat{w}'_j = P_u$ for ReLU, since the functions $|xu|$ of distinct classes are linearly independent (Step 2 of Theorem A.4), and for GELU, by Lemmas A.5 and A.6, $\sum_{|s_j| = \alpha} \hat{w}'_j = \hat{w}_i$ if w_i lies in the class and $\alpha = \|w_i\|$, and 0 otherwise. With $Q_u = \sum_j s_j \hat{w}'_j$ the odd part reads $\sum_u u\, (Q_u - P_u) = 2M$. Call a class mixed if it contains w'_j of both signs, for GELU of both signs and equal $|s_j|$. A class that is not mixed has $Q_u = P_u$ or $Q_u = -P_u$. For ReLU all its s_j have one sign. For GELU all s_j of one scale α have one sign ϵ_α, so $Q_u = \sum_\alpha \epsilon_\alpha\, \alpha \sum_{|s_j| = \alpha} \hat{w}'_j = \pm \|w_i\| \hat{w}_i$ in the class of w_i, and $Q_u = 0 = P_u$ in a class without a column of W_{read}. Let S collect the units i whose class is not mixed and has $Q_u = -P_u$. For $i \in S$, $u\, (Q_u - P_u) = -2\, w_i \hat{w}_i$, so the classes that are not mixed contribute $-2\, W_{\text{read}}[:, S]\, W_{\text{write}}[S, :]$ to the sum, and

$$2\big(M + W_{\text{read}}[:, S] W_{\text{write}}[S, :]\big) = \sum_{u \text{ mixed}} u\, (Q_u - P_u).$$

Every class of a column w_i contains some w'_j, since $P_u \neq 0$, and every mixed class contains at least two, so $m' \geq m$ plus the number of mixed classes. If $m' < m + d_{\text{model}}$, the right side has rank less than d_{model}, so the left side is singular, which the weights exclude. □

Theorem 3.3 (No width suffices)**.** *Let B_θ be (a) a block that is positively homogeneous of degree $k \neq 1$, (b) a gated block whose gate vanishes at 0 and is differentiable there, or (c) a $\tanh$ block. ReLU^2 and ReGLU blocks fall under (a), SwiGLU and GeGLU blocks under (b). For every $M \neq 0$, no block $B_{\theta'}$ of the same kind and of any width satisfies (2).*

Theorem 4.1 (No equality for ReLU^2, gated and $\tanh$ networks)**.** *Let the residual network have $ED \neq 0$.*
(i) *Without normalization or with pre-RMSNorm and $\varepsilon > 0$, let the blocks be gated blocks whose gate vanishes at 0 and is differentiable there, or blocks that are positively homogeneous of degree $k > 1$. No skipless network of blocks of the same kind, of any depth and widths, equals the residual network.*
(ii) *Without normalization, no skipless network of $\tanh$ blocks, of any depth and widths, equals a residual network of $\tanh$ blocks.*

Proof of Theorems 3.3 and 4.1. For homogeneous and gated blocks, Theorem 3.3 is Propositions A.1 and A.2, and Theorem 4.1(i) is Corollary A.3. For $\tanh$ blocks, B_θ and $B_{\theta'}$ are bounded and xM is not, which gives Theorem 3.3, and Theorem 4.1(ii) is again Corollary A.3. □

Theorem 3.1 (Absorption with d_{model} more units)**.** *Let σ be ReLU or GELU, and fix a matrix $M \neq 0$.*
(i) *If B_θ is nondegenerate, some block $B_{\theta'}$ of the same width satisfies (2) if and only if some set S of hidden units has $W_{\text{read}}[:, S]\, W_{\text{write}}[S, :] = -M$. For generic θ, no block of the same width does.*
(ii) *If M is invertible and W_{read} has rank d_{model}, some block $B_{\theta'}$ with d_{model} more hidden units satisfies (2), and for generic θ none with fewer does.*

Proof of Theorem 3.1. Part (i) is Corollary A.9 with $p = q = d_{\text{model}}$, which for $M = I$ is Theorems A.4 and A.7; generic blocks are nondegenerate, and for each $M \neq 0$ the condition holds only on a set of weights of measure zero. Part (ii) is Proposition A.11. □

Proposition 3.2 (Networks of one block)**.** *Let σ be ReLU or GELU. For almost every choice of weights, a residual network of one block equals no skipless network of one block of the same width.*

Proof of Proposition 3.2. A residual network of one block and a skipless network of one block of the same width are the maps

$$X \mapsto XED + f(XEW_{\text{read}})\, W_{\text{write}} D \qquad \text{and} \qquad X \mapsto f(XE'W'_{\text{read}})\, W'_{\text{write}} D'.$$

Both act on each row of X separately. Apply Corollary A.9 with $p = d_{\text{in}}$, $q = d_{\text{out}}$, the reads EW_{read}, the writes $W_{\text{write}}D$ and $M = ED$. For generic E, W_{read}, W_{write} and D, the columns of EW_{read} are nonzero and pairwise non-collinear and the rows of $W_{\text{write}}D$ are nonzero. The two networks are then equal only if $EW_{\text{read}}[:, S]\, W_{\text{write}}[S, :]\, D = -ED$ for some set S of units. For each S this is a system of polynomial equations in the weights that fails at $W_{\text{write}} = 0$ when $ED \neq 0$, so for generic weights it does not hold. □

Theorem 4.5 (The first block needs $2d_{\text{model}}$ units)**.** *Consider networks of ReLU blocks, or of GELU blocks, without normalization. For almost every choice of weights, a residual network with hidden dimension d_{model} equals no skipless network whose first block has fewer than $2d_{\text{model}}$ hidden units, whatever its depth and its other widths.*

Proof of Theorem 4.5. MLP blocks act on each token separately, so we compare the networks on rows $x \in \mathbb{R}^{1 \times d_{\text{in}}}$. Outside a set of weights of measure zero, E and D have rank d_{model}, no read of a unit vanishes, and $I + W^\ell_{\text{read}}[:, S]\, W^\ell_{\text{write}}[S, :]$ is invertible for every block ℓ and every set S of its units, since each determinant is a polynomial in the weights that equals 1 when the writes vanish. A unit is *active* at an input if its argument there is positive, and it *switches* at an input if its argument vanishes there. On an open set of inputs on which no unit of the residual network switches, with active

sets S_ℓ, the network is the linear map $x \mapsto xE\Theta D$ with $\Theta = \prod_\ell \left(I + W^\ell_{\text{read}}[:, S_\ell]\, W^\ell_{\text{write}}[S_\ell, :]\right)$ invertible, of rank d_{model}. The inputs at which a unit switches lie in finitely many hyperplanes, by induction over the blocks: off the hyperplanes of blocks $1, \dots, \ell - 1$, the active sets of these blocks take finitely many values, and for each of them the argument of a unit of block ℓ is x times a vector $E\Theta' w$, with Θ' invertible and $w \neq 0$ the read of the unit. So the inputs near which the network is linear of rank d_{model} form an open dense set Ω.

Let a skipless network F' have embedding E' and a first block with reads $w'_1, \dots, w'_{m'}$, $m' < 2d_{\text{model}}$, and set $a_i = E' w'_i$. Take x_0 with $x_0, -x_0 \in \Omega$ and $x_0 a_i \neq 0$ whenever $a_i \neq 0$. The units active near x_0, those with $x_0 a_i > 0$, and those active near $-x_0$ are disjoint, so one of the two sets, say the first after replacing x_0 by $-x_0$ if needed, has fewer than d_{model} units. Near x_0 the other units of the first block output zero, so F' depends on x only through the numbers xa_i of the active units, and $F'(x_0 + u) = F'(x_0)$ for every small u in a subspace of dimension greater than $d_{\text{in}} - d_{\text{model}}$. The residual network F changes by $uE\Theta D$, which vanishes only on a subspace of dimension $d_{\text{in}} - d_{\text{model}}$. So some small u changes F and leaves F' unchanged, and $F' \neq F$.

GELU. As $\text{GELU}(y) - \text{ReLU}(y) = -|y|\, \Phi(-|y|)$, with Φ the standard normal distribution function, we have $|\text{GELU}(ty)/t - \text{ReLU}(y)| \leq \kappa/t$ for all $y \in \mathbb{R}$ and $t > 0$, where $\kappa = \max_{u \geq 0} u\, \Phi(-u) < 0.17$. Given a network of GELU blocks without normalization, let $h_\ell(t)$ be its hidden states at the input tx, and h^{R}_ℓ those of the ReLU network with the same weights at the input x. Then $h_0(t)/t = h^{\text{R}}_0$. If $h_{\ell-1}(t)/t = h^{\text{R}}_{\ell-1} + O(1/t)$, the output of block ℓ, divided by t, is $\text{GELU}\big(h_{\ell-1}(t) W^\ell_{\text{read}}\big) W^\ell_{\text{write}}/t = \text{ReLU}\big(h^{\text{R}}_{\ell-1} W^\ell_{\text{read}}\big) W^\ell_{\text{write}} + O(1/t)$, by the bound and since ReLU is 1-Lipschitz, so $h_\ell(t)/t = h^{\text{R}}_\ell + O(1/t)$, with or without the skip. So the output of the GELU network at tx, divided by t, tends to the output of the ReLU network at x. If a residual GELU network equals a skipless one, the two ReLU networks with the same weights are therefore equal. By the case of ReLU, the weights of the residual network for which this happens with a skipless network whose first block has fewer than $2d_{\text{model}}$ units form a set of measure zero. □

Proposition 4.4 (d_{model} more units per block suffice)**.** *Let σ be ReLU or GELU. Without normalization, a residual network whose reads all have rank d_{model} equals a skipless network with d_{model} more hidden units in every block.*

Proof of Proposition 4.4. Theorem 3.1(ii) with $M = I$ turns each residual block, whose read has rank d_{model}, into a skipless block with d_{model} more hidden units. Composing these blocks, with the same embedding and readout, gives a skipless network that equals the residual one. □

Proposition 4.6 (Injective blocks: $2d_{\text{model}}$ units in every block)**.** *Let σ be ReLU or GELU, without normalization, and let E and D have rank d_{model}. Suppose that every block of the residual network is injective on tokens when σ is replaced by ReLU: $x + B_\ell(x) \neq x' + B_\ell(x')$ for all tokens $x \neq x'$. Then every skipless network of σ blocks, of any depth and with the same d_{model}, that equals the residual network has at least $2d_{\text{model}}$ hidden units in every block.*

Proof of Proposition 4.6. As in the proof of Theorem 4.5, we compare the networks on rows $x \in \mathbb{R}^{1 \times d_{\text{in}}}$, and GELU reduces to ReLU: if a residual GELU network equals a skipless one, the two ReLU networks with the same weights are equal, with the same widths, and the hypothesis is about this residual ReLU network. So let σ be ReLU. Every block below is positively homogeneous, $B(ch) = cB(h)$ for $c > 0$, since there are no biases.

Step 1: the residual network is injective. Let $\Phi_\ell(h) = h + B_\ell(h)$. By hypothesis each Φ_ℓ is injective. As D has rank d_{model}, the residual network F satisfies $F(x) = F(x')$ only if $xE = x'E$.

Step 2: every skipless block is injective. Let a skipless network $F' = F$ have embedding E', blocks $B'_1, \dots, B'_{L'}$ and readout D', and write $F'(x) = H(xE')$. If $xE' = x'E'$, then $F(x) = F(x')$, so $xE = x'E$: the kernel of E' lies in that of E. As E has rank d_{model}, so does E', the two kernels are equal, and H is injective on all of $\mathbb{R}^{1 \times d_{\text{model}}}$. In particular B'_1 is injective. A continuous, positively homogeneous, injective map ψ of $\mathbb{R}^{1 \times d_{\text{model}}}$ to itself is onto: $\psi(0) = 0$ by homogeneity, so $\psi(u) \neq 0$ for $u \neq 0$, so $\|\psi(u)\| \geq c\|u\|$ with $c > 0$ the minimum of $\|\psi\|$ on the unit sphere, hence the image of ψ is closed; by invariance of domain it is open; and it is not empty, so it is all of the connected space $\mathbb{R}^{1 \times d_{\text{model}}}$. So B'_1 is onto, B'_2 is injective on all of $\mathbb{R}^{1 \times d_{\text{model}}}$, and by induction every block B'_k is.

Step 3: an injective block has $2d_{\text{model}}$ units. Let B'_k have reads $r_1, \dots, r_{m'}$, and drop the units with $r_i = 0$, which are never active. Fix u off the hyperplanes $\{u : ur_i = 0\}$; as they pass through 0, so is $-u$. Near each of the two points the block is linear, equal to $v \mapsto v\, W'_{\text{read}}[:, S]\, W'_{\text{write}}[S, :]$ with S the set of units active there. Injectivity makes this matrix invertible, so $|S| \geq d_{\text{model}}$ at u and at $-u$. The active sets at u and $-u$ are disjoint, so B'_k has at least $2d_{\text{model}}$ units. □

Remark A.12 (A first block that is not injective). For ReLU, the conclusion of Proposition 4.6 can fail when a block is not injective. Let E and D have rank d_{model}, let block 1 have d_{model} units with $W_{\text{read}} = -I$ and $W_{\text{write}} = 2I$, and let blocks $2, \dots, L$ have d_{model} units each, with reads whose entries are all negative. Then $h + B_1(h) = (|h_1|, \dots, |h_{d_{\text{model}}}|)$, so $(h + B_1(h))\, r < 0$ for every $h \neq 0$ and every read r of a later block. By homogeneity, this says that a maximum over the unit sphere is negative, so it holds on an open set of weights. On this set the units of later blocks are never active, and the residual network computes $x \mapsto \big(xE + B_1(xE)\big)D$. Let K be the closed convex cone spanned by the image of $h \mapsto h + B_1(h)$; every later read is negative on $K \setminus \{0\}$, so K is pointed. A skipless network equals the residual one with $2d_{\text{model}}$ units in its first block (Theorem 3.1(ii), as the read of block 1 has rank d_{model}) and d_{model} units in each later block: these compute the identity on K with linearly independent reads $v_1, \dots, v_{d_{\text{model}}}$, positive on $K \setminus \{0\}$, and the writes $[\, v_1 \cdots v_{d_{\text{model}}} \,]^{-1}$. So the later blocks of this skipless network have fewer than $2d_{\text{model}}$ units.

Proposition 4.7 (A lower bound from counting dimensions)**.** *Consider ReLU networks of L blocks without normalization. The full residual networks form an open set of weights. For almost every full residual network, every skipless network of L blocks with the same d_{model} that equals it has, in total, at least $Ld_{\text{model}}^2/(2d_{\text{model}} - 1)$ more hidden units.*

Proof of Proposition 4.7. We compare the networks on rows $x \in \mathbb{R}^{1 \times d_{\text{in}}}$. For a finite set Z of rows, let $F_Z(\theta) \in \mathbb{R}^{|Z| d_{\text{out}}}$ collect the outputs of the residual network with weights $\theta \in \mathbb{R}^P$ at the inputs of Z, and let $r = P - d_{\text{model}}^2 - s$.

Step 1: patterns give polynomial maps. A *pattern* on Z labels every unit, at every input of Z, active or inactive. Given a pattern π, replace $\mathrm{ReLU}(z)$ in the network by z where π says active and by 0 where it says inactive. The outputs on Z become a polynomial map F_Z^π of the weights. The pattern of θ labels a unit active at an input if its argument there is positive, and inactive otherwise; with this pattern, $F_Z(\theta) = F_Z^\pi(\theta)$.

Step 2: the rank of a pattern map. The map F_Z^π does not change when a change of basis G acts on all hidden states, $(E, W_{\text{read}}^\ell, W_{\text{write}}^\ell, D) \mapsto (EG, G^{-1}W_{\text{read}}^\ell, W_{\text{write}}^\ell G, G^{-1}D)$, nor when the read of a unit is multiplied by $c > 0$ and its write by $1/c$, since these scalings commute with the labels. Differentiating along these families at $G = I$ and $c = 1$ gives vectors in the kernel of the Jacobian of F_Z^π at θ: for every $d_{\text{model}} \times d_{\text{model}}$ matrix A, the vector with the entries $(EA, -AW_{\text{read}}^\ell, W_{\text{write}}^\ell A, -AD)$, and for every unit, the vector that holds its read and minus its write. If a combination of them vanishes, its entry EA vanishes, so $A = 0$ when E has rank d_{model}; the entries at the reads then show that every unit with a nonzero read has coefficient 0. So for generic θ these $d_{\text{model}}^2 + s$ vectors are linearly independent, and the Jacobian of F_Z^π has rank at most r there. A polynomial map reaches its largest rank on a nonempty open set, so the rank is at most r at every θ.

The same argument applies to a skipless network of L blocks with the same d_{model} and n more hidden units in total. It has $P + 2d_{\text{model}}n$ weights, and each of its blocks has at least d_{model} units (Section 2), so for generic weights its writes have rank d_{model}. Its pattern maps $F_Z'^\pi$ do not change under a change of basis at each of its $L + 1$ hidden states, nor under the rescalings of its $s + n$ units. The vectors obtained by differentiating are, for a matrix A_k at hidden state k, the one with the entries $W_{\text{write}}^k A_k$ and $-A_k W_{\text{read}}^{k+1}$, with E in place of W_{write}^0 and D in place of W_{read}^{L+1}, and one vector per unit as above. They are linearly independent when E and every write have rank d_{model} and no read of a unit vanishes: in a vanishing combination, the entry at E gives $A_0 = 0$, the entries at the reads of block 1 then give coefficient 0 for its units, the entry at its write gives $A_1 = 0$, and so on. So every $F_Z'^\pi$ has rank at most $P + 2d_{\text{model}}n - (L + 1)d_{\text{model}}^2 - s - n$ at every point.

Step 3: full networks form an open set. Let θ_0 be full, with the set Z. No argument vanishes at θ_0 on Z, so the pattern on Z is constant near θ_0, and F_Z is there a polynomial map of rank at most r, by Step 2, and of rank r at θ_0. The rank is lower semicontinuous, so it is r on a neighborhood V of θ_0, and every network of V is full.

Step 4: almost every network of V equals no skipless network. By the constant rank theorem, after shrinking V, F_Z maps V onto an embedded manifold Y of dimension r, and a linear projection pr onto $\mathbb{R}^r$ maps Y diffeomorphically onto an open set. Let $n < Ld_{\mathrm{model}}^2/(2d_{\mathrm{model}} - 1)$, so that every map $F_Z'^{\pi}$ of a skipless network with n more units has rank less than r. Then all values of $\mathrm{pr} \circ F_Z'^{\pi}$ are critical, and by Sard's theorem its image is a null set of $\mathbb{R}^r$. If the residual network with weights $\theta \in V$ equals a skipless network with weights η, then $F_Z(\theta) = F_Z'^{\pi}(\eta)$ for the pattern π of η, so $\mathrm{pr}(F_Z(\theta))$ lies in this null set. As $\mathrm{pr} \circ F_Z$ is a submersion on V, such θ form a null set. There are finitely many n below the bound, finitely many choices of widths with n more units and finitely many patterns on Z, and countably many neighborhoods V cover the full networks. □

A.2 No absorption in attention (Theorems 3.4 and 4.2, Proposition A.15)

Theorem 3.4 collects what the main text uses of the following theorem, for the attention blocks B_θ of (4).

Theorem A.13 (Attention and linear terms)**.** *Let θ and θ' be parameters of attention blocks with any numbers of heads and any head dimensions, and let $M \neq 0$ be a $d_{\mathrm{model}} \times d_{\mathrm{model}}$ matrix.*
(i) *For almost every $h \in \mathbb{R}^{T \times d_{\mathrm{model}}}$, $hM + B_\theta(h) \neq B_{\theta'}(h)$.*
(ii) *Some h whose rows have norm at most 1 has a row in which $hM + B_\theta(h)$ and $B_{\theta'}(h)$ differ by at least $\|M\|/3$, where $\|M\|$ is the spectral norm.*
(iii) *With an embedding E and a readout D with $ED \neq 0$, the residual network $X \mapsto \big(XE + B_\theta(XE)\big)D$ differs from every skipless network $X \mapsto B_{\theta'}(XE')D'$.*

Overview. Two kinds of input suffice. If all tokens are equal, attention returns them unchanged, so the value–output products of all heads must add up to M. If one token is zero, all its scores are zero, so every head gives it the average of the tokens, while the skip gives it zero; this asks the same sum to vanish. Part (ii) makes the conflict quantitative, and analyticity turns it into part (i). With an embedding and a readout, the same two inputs give part (iii).

Proof of Theorem A.13. Let $F(h) = hM + B_\theta(h) - B_{\theta'}(h)$, and let θ' have H' heads. The difference $B_{\theta'} - B_\theta$ is one attention block with $H + H'$ heads,

$$B_{\theta'}(h) - B_\theta(h) = \sum_{k=1}^{H+H'} R_k(h)\, h\, C_k,$$

where the $R_k(h)$ are the attention maps of the heads of θ' and of θ, and the C_k are their value–output products, with a minus sign for the heads of θ. Let $C = \sum_k C_k$.

Step 1: equal tokens. Let every row of h be x. Each row of $R_k(h)$ sums to 1, so every row of $R_k(h)h$ is x. Hence every row of $F(h)$ is $x(M - C)$.

Step 2: one zero token. Let the last row of h be 0 and the other rows y. Every score in the last row of $R_k(h)$ is a product with the last row of h, which is 0. So that row of $R_k(h)$ is $(1/T, \ldots, 1/T)$, also with a causal mask, since the last token sees all T tokens. Hence the last row of $F(h)$ is $-\frac{T-1}{T}\, yC$.

Proof of (ii). Over $\|x\|, \|y\| \leq 1$, Step 1 gives rows of norm up to $\|M - C\|$, and Step 2 up to $\frac{T-1}{T}\|C\| \geq \frac{1}{2}\|C\|$. If $\|M - C\| < \|M\|/3$, then $\|C\| > \frac{2}{3}\|M\|$, so Step 2 reaches more than $\|M\|/3$.

Proof of (i). F is real-analytic, because the softmax is, and by (ii) some entry of F is not identically zero. The zeros of a real-analytic function that is not identically zero form a set of measure zero: in one variable they are isolated, and the general case follows by induction on the dimension with Fubini's theorem.

Proof of (iii). Let C_θ and $C_{\theta'}$ be the sums of the value–output products of the heads of θ and of θ'. On inputs whose rows all equal a row vector u, Step 1 gives the rows $uE(I + C_\theta)D$ in the residual network and $uE'C_{\theta'}D'$ in the skipless network. On inputs whose last row is 0 and whose other rows equal v, Step 2 gives the last rows $\frac{T-1}{T}\, vEC_\theta D$ and $\frac{T-1}{T}\, vE'C_{\theta'}D'$. If the two networks are equal, then $E(I + C_\theta)D = E'C_{\theta'}D' = EC_\theta D$, so $ED = 0$. □

We recall Theorem 3.4 from Section 3.

Theorem 3.4 (No absorption in attention blocks)**.** *Let θ and θ' be parameters of attention blocks with any numbers of heads and any head dimensions, and let $M \neq 0$ be a $d_{\text{model}} \times d_{\text{model}}$ matrix.*
(i) *For almost every $h \in \mathbb{R}^{T \times d_{\text{model}}}$, $hM + B_\theta(h) \neq B_{\theta'}(h)$.*
(ii) *Some h whose rows have norm at most 1 has a row in which $hM + B_\theta(h)$ and $B_{\theta'}(h)$ differ by at least $\|M\|/3$, where $\|M\|$ is the spectral norm.*

Proof of Theorem 3.4. Its two parts are parts (i) and (ii) of Theorem A.13. □

Proposition A.15 follows from a lemma on attention heads.

Lemma A.14 (Attention heads are identifiable)**.** *Let $T \geq 3$. Let $\Gamma_1, \dots, \Gamma_m \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$ be distinct, let $C_1, \dots, C_m, M \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$, and write $P_\Gamma(h) = \operatorname{softmax}(h\Gamma h^\top)$, with the softmax taken row by row, with or without a causal mask. If*

$$hM + \sum_{k=1}^{m} P_{\Gamma_k}(h)\, h\, C_k = 0$$

for all h in a nonempty open subset of $\mathbb{R}^{T \times d_{\text{model}}}$, then $M = 0$ and $C_1 = \cdots = C_m = 0$.

Overview. Move one token along a complex line, and watch the last token, which sees every token under either mask. Each head becomes a meromorphic function of one complex variable, whose poles lie on a vertical line that its score matrix determines, so heads with distinct score matrices have disjoint poles. At a pole of one head the other terms are regular, so its residue must vanish, and the residues force its value–output product to vanish. What is left is the skip and a uniform head, which two rows of one input separate.

Proof of Lemma A.14. One moving token. Fix h in the open set, with rows $x_1, \dots, x_T$, and fix $v \in \mathbb{R}^{1 \times d_{\text{model}}}$. Replace the second row by $x_2 + tv$ with $t \in \mathbb{C}$, and look at the last row, which attends to all T tokens with or without a causal mask; as $T \geq 3$, it is not the second row. For a head with score matrix Γ, the last row has the scores $z_r = x_T \Gamma x_r^\top$ for $r \neq 2$, and $a + bt$ for $r = 2$, where $a = x_T \Gamma x_2^\top$ and $b = x_T \Gamma v^\top$. Its output is

$$g_\Gamma(t) = \frac{e^{a+bt}\,(x_2 + tv) + Z\bar{x}}{e^{a+bt} + Z}, \qquad Z = \sum_{r \neq 2} e^{z_r}, \qquad \bar{x} = \frac{1}{Z} \sum_{r \neq 2} e^{z_r} x_r,$$

where $\bar{x}$ is the average of the other tokens under the weights of the head. The last row of hM is $x_T M$, which does not depend on t.

Poles. If $b \neq 0$, then g_Γ is meromorphic on $\mathbb{C}$, with simple poles at

$$t_n = c + \frac{(2n+1)\pi i}{b}, \quad n \in \mathbb{Z}, \qquad \text{where } c = \frac{\log Z - a}{b},$$

so all its poles lie on the vertical line $\operatorname{Re} t = c$. The residue at t_n is $(x_2 + t_n v - \bar{x})/b$. If $\Gamma = 0$, then g_Γ is affine in t and has no poles.

The poles tell the heads apart. For h and v outside a closed set with empty interior, $b \neq 0$ for every $\Gamma_k \neq 0$, and distinct $\Gamma_k \neq 0$ give distinct lines $\operatorname{Re} t = c$. The first condition fails only on the zero sets of nonzero polynomials. For the second, suppose that two heads $\Gamma \neq \Gamma'$ have $c = c'$ for all h and v near some point, that is $(\log Z - a)\, b' = (\log Z' - a')\, b$, where Z and Z' do not depend on x_2 or v. Comparing the terms bilinear in (x_2, v) gives $x_T \Gamma' = \lambda\, x_T \Gamma$ for some number $\lambda \neq 0$. Comparing the terms linear in v then gives $\log Z' = \lambda \log Z$, that is,

$$\log \sum_{r \neq 2} e^{\lambda z_r} \;=\; \lambda \log \sum_{r \neq 2} e^{z_r}.$$

Because $T \geq 3$, the sum has at least two terms, $r = 1$ and $r = T$, and z_1 varies with x_1 while λ and the other z_r stay fixed. Differentiating in z_1 gives $\operatorname{softmax}(\lambda z)_1 = \operatorname{softmax}(z)_1$ on an interval of

z_1, which forces $\lambda = 1$. So $x_T\Gamma' = x_T\Gamma$ for all x_T near a point, and $\Gamma' = \Gamma$, a contradiction. With two tokens this step fails, as the example after Proposition A.15 shows.

Residues. The identity holds for real t near 0, so by analytic continuation it holds on all of $\mathbb{C}$ away from the poles. At a pole t_n of a head $\Gamma_k \neq 0$, no other term is singular, so the residue times C_k vanishes: $(x_2 + t_n v - \bar{x})\, C_k = 0$ for every $n \in \mathbb{Z}$. Subtracting two consecutive ones gives $vC_k = 0$. This holds for all v in an open set, so $C_k = 0$.

The skip. What remains is $hM + P_0(h)\, h\, C_0 = 0$, where C_0 is the coefficient of $\Gamma = 0$ (or 0 if no Γ_k vanishes). This is linear in h, so it holds for all h. Take the first row of h equal to x and the other rows 0, and let T_i be the number of tokens that token i attends to: $T_1 = T_2 = T$ without a mask, and $T_1 = 1$, $T_2 = 2$ with a causal mask. The first two rows give $x(M + C_0/T_1) = 0$ and $xC_0/T_2 = 0$. So $C_0 = 0$ and $M = 0$. □

The lemma gives the proposition that Section 3.3 cites.

Proposition A.15 (The softmaxes must match)**.** *Let $T \geq 3$, let θ and θ' be parameters of attention blocks, and suppose that $hM + B_\theta(h) = B_{\theta'}(h)$ for all h in a nonempty open set. Then $M = 0$, and for every matrix Γ, the heads with score matrix Γ have the same sum of value–output products in B_θ as in $B_{\theta'}$.*

Proof of Proposition A.15. Group the heads of $B_{\theta'}$ and B_θ by score matrix, add their value–output products, with a minus sign for the heads of θ, and apply Lemma A.14. □

With two tokens, heads can cancel: the softmax of two scores is the logistic function of their difference, and $1/(1 + e^{-z}) + 1/(1 + e^{z}) = 1$ makes two heads with opposite score matrices add up to a uniform head. A skip still cannot be absorbed when $T = 2$ (Theorem 3.4).

Theorem 4.2 (Attention networks on long and short sequences)**.** *Without normalization or with pre-RMSNorm and $\varepsilon > 0$, let a residual network of L attention blocks have $ED \neq 0$. Consider skipless networks of $L' \geq 1$ attention blocks, with any numbers of heads and any head dimensions.*
(i) *If the attention has no mask, no such skipless network equals the residual one. With a causal mask on T tokens, the same holds when $T \geq L + 2$, and when $T = L + 1$ and $L' \leq L$.*
(ii) *For almost every choice of weights of the residual network, no such skipless network equals it, whatever L'.*
(iii) Short sequences. *Without normalization and with a causal mask, in every other case, $T \leq L$ or $T = L + 1 \leq L'$, some residual network of L blocks with $ED \neq 0$ equals a skipless network of L' blocks.*

Proof. Let $\xi = 1$ without normalization and $\xi = 1/\sqrt{\varepsilon}$ with pre-RMSNorm, so that $N(h) = \xi h + O(\|h\|^3)$ near $h = 0$ in the second case. The scores $h\Gamma_i h^\top$ of an attention block are quadratic in h, so $P_{\Gamma_i}(h) = U + O(\|h\|^2)$, where $U = \mathrm{softmax}(0)$ is the $T \times T$ matrix whose row t is uniform on the tokens that token t attends to: all T tokens without a mask, the first t tokens with a causal mask. Hence $B_\theta(h) = UhV + O(\|h\|^3)$, where $V = \sum_i W_{V,i} W_{O,i}$ is the sum of the value–output products of the heads.

Every map below fixes 0, so the Jacobians at 0 compose. Let the blocks of the skipless network have the sums $V'_1, \ldots, V'_{L'}$, and those of the residual network $V_1, \ldots, V_L$. At $X = 0$ the skipless network has the Jacobian $\delta \mapsto U^{L'} \delta\, \Lambda'$, with $\Lambda' = \xi^{L'+1} E' V'_1 \cdots V'_{L'} D'$. Block ℓ of the residual network has the Jacobian $\delta \mapsto \delta + \xi\, U \delta V_\ell$ at 0. Multiplying by U on the left commutes with multiplying by V_ℓ on the right, so the residual network has the Jacobian

$$\delta \mapsto \sum_{k=0}^{L} U^k \delta\, \Lambda_k, \qquad \Lambda_k = \xi^{k+1} E \Big(\sum_{\ell_1 < \cdots < \ell_k} V_{\ell_1} \cdots V_{\ell_k} \Big) D, \qquad \text{so that } \Lambda_0 = \xi\, ED.$$

If the two networks are equal, their Jacobians at $X = 0$ are equal:

$$\sum_{k=0}^{L} U^k \delta\, \Lambda_k = U^{L'} \delta\, \Lambda' \qquad \text{for all } \delta \in \mathbb{R}^{T \times d_{\text{in}}}. \tag{9}$$

Proof of (i), no mask. Every entry of U is $1/T$. Let $a \in \mathbb{R}^{T\times 1}$ be a nonzero column whose entries add up to 0, so that $Ua = 0$, and let $\delta = ab$ for a row $b \in \mathbb{R}^{1\times d_{\text{in}}}$. As $L' \geq 1$, (9) reads $a\, b\Lambda_0 = 0$. So $bED = 0$ for every b, that is $ED = 0$, a contradiction.

Proof of (i), causal mask. Now U is lower triangular with the diagonal entries $1, 1/2, \ldots, 1/T$. These are T distinct eigenvalues; let a_t be a column with $Ua_t = a_t/t$. With $\delta = a_t b$, (9) gives $\sum_k t^{-k}\, b\Lambda_k = t^{-L'}\, b\Lambda'$ for every row b. So the matrix polynomial

$$p(z) = \sum_{k=0}^{L} \Lambda_k z^k - \Lambda' z^{L'}$$

vanishes at the T points $z = 1, 1/2, \ldots, 1/T$. If $L' \leq L$, every entry of p is a polynomial of degree at most L with $T \geq L+1$ roots, so $p = 0$. If $L' > L$, the hypothesis gives $T \geq L+2$, and every entry of p has at most $L+2$ nonzero coefficients. By Descartes' rule of signs, a real polynomial with n nonzero coefficients that is not zero has at most $n-1$ positive roots. So again $p = 0$. In both cases the constant coefficient of p vanishes. As $L' \geq 1$, it is $\Lambda_0 = \xi\, ED \neq 0$, a contradiction.

Proof of (ii). By (i), only causal masks remain. From $p(1) = 0$ and $p(1/2) = 0$, $\sum_{k=0}^{L} \big(2^{-k} - 2^{-L'}\big)\Lambda_k = 0$. The left side is a polynomial in the weights of the residual network. When all value–output products vanish it equals $(1 - 2^{-L'})\xi\, ED$, so it is not the zero polynomial, and it vanishes only on a null set of weights. The union of these null sets over $L' \geq 1$ is a null set.

Proof of (iii). Let $z_t = 1/t$ for $t = 1, \ldots, T$. We first choose real numbers $q_1, \ldots, q_L$ and γ' such that the polynomial $Q(z) = 1 + q_1 z + \cdots + q_L z^L$ satisfies

$$Q(z_t) = \gamma' z_t^{L'} \qquad \text{for } t = 1, \ldots, T. \tag{10}$$

If $T \leq L$, let $\gamma' = 1$. The $T \times L$ matrix with the entries z_t^k, $1 \leq k \leq L$, has rank T, since its first T columns form a Vandermonde matrix with its rows multiplied by the z_t. So the linear equations (10) in $q_1, \ldots, q_L$ have a solution. If $T = L+1 \leq L'$, (10) is a system of $L+1$ linear equations in $q_1, \ldots, q_L$ and γ'. As $L' > L$, the monomials $z, \ldots, z^L, z^{L'}$ are distinct, so a vector in the kernel of its matrix gives a polynomial with at most $L+1$ nonzero coefficients and the $L+1$ positive roots z_t, which is zero by Descartes' rule. So the matrix is invertible, and there is a solution.

Next we realize Q by L blocks. As $Q(0) = 1$, the real polynomial Q is a product of factors $1 + az$ and $1 + 2\alpha z + (\alpha^2 + \beta^2)z^2$ with real a, α and β. Let $\Pi = \operatorname{diag}(1, 1, 0, \ldots, 0)$, and let the $d_{\text{model}} \times d_{\text{model}}$ matrix J have the entries $J_{12} = -1$ and $J_{21} = 1$ and zeros elsewhere, so that $J^2 = -\Pi$ and $J\Pi = \Pi J = J$. Then

$$I + az\,\Pi = (1 + az)\,\Pi + (I - \Pi),$$
$$\big(I + z(\alpha\Pi + \beta J)\big)\big(I + z(\alpha\Pi - \beta J)\big) = \big(1 + 2\alpha z + (\alpha^2 + \beta^2)z^2\big)\,\Pi + (I - \Pi).$$

Give the residual network one block with $V_\ell = a\Pi$ for each factor of the first kind, two blocks with $V_\ell = \alpha\Pi + \beta J$ and $V_\ell = \alpha\Pi - \beta J$ for each factor of the second kind, and $V_\ell = 0$ in its other blocks; this takes at most L blocks, as Q has degree at most L. All these matrices commute, and

$$\prod_{\ell=1}^{L} (I + zV_\ell) = Q(z)\,\Pi + (I - \Pi).$$

Let every score matrix vanish, in both networks. Then every attention matrix is U, block ℓ is the linear map $h \mapsto h + UhV_\ell$, and the residual network is $X \mapsto \sum_k U^k X \Lambda_k$, where $\sum_k \Lambda_k z^k = E\prod_\ell (I + zV_\ell)D$. Let E and D have the entries $E_{11} = E_{22} = D_{11} = D_{22} = 1$ and zeros elsewhere. Then $ED \neq 0$ and $E\Pi = E$, so $\sum_k \Lambda_k z^k = Q(z)\, ED$, and the residual network is $X \mapsto Q(U)\, XED$. The matrix U is diagonalizable with the eigenvalues z_t, so (10) gives $Q(U) = \gamma' U^{L'}$. The skipless network with the same E and D whose blocks have $V_1' = \gamma'\Pi$ and $V_\ell' = \Pi$ for $\ell \geq 2$ is $X \mapsto \gamma' U^{L'} XED$, so the two networks are equal. A block has the sum V once its heads have value–output products that add up to V, which two heads of any dimension allow for matrices of rank at most 2.

Parts (i) and (ii) use only that the attention weights are uniform when all tokens vanish, so they also hold with rotary embeddings. □

A.3 Networks under RMSNorm (Theorem 4.3)

We recall Theorem 4.3 from Section 4.

Theorem 4.3 (Almost no equality under RMSNorm)**.** *Consider pre-RMSNorm networks, with $\varepsilon \geq 0$ and blocks of any kind and width. For almost every choice of weights, a residual network equals no skipless network, whatever its depth, widths and kinds of blocks.*

Write N_0 for RMSNorm with $\varepsilon = 0$, which maps each nonzero row w to $\sqrt{d_{\text{model}}}\, w/\|w\|$, with the derivative $\mathrm{D}N_0(w)[b] = \sqrt{d_{\text{model}}}\,\big(b - (b \cdot u)\, u\big)/\|w\|$, where $u = w/\|w\|$; it vanishes exactly when b is a multiple of w. Outside a set of weights of measure zero, E and D have rank d_{model}, so that XE takes every value in $\mathbb{R}^{T \times d_{\text{model}}}$ and $vD = 0$ only for $v = 0$. Each block is linear in its write, which for an attention block holds the matrices $W_{O,i}$: block ℓ adds $a_\ell W^\ell_{\text{write}}$ to the hidden state, where a_ℓ, a function of the normalized state n that block ℓ reads, holds the activations of its hidden units, or for an attention block the matrices $P_{\Gamma_i}(n)\, n\, W_{V,i}$ of its heads side by side. It depends on the read of block ℓ only.

Proof of Theorem 4.3. Fix an input X for which XE has no zero row, and let $t \to \infty$.

The skipless network. Its hidden states depend on the input only through $N(tXE')$, at its embedding E'. On a row w of XE', $N(tw) = w/\sqrt{\|w\|^2/d_{\text{model}} + \varepsilon/t^2}$. For $\varepsilon = 0$ this is $N(w)$, so the output at tX is the output at X. For $\varepsilon > 0$ it is a smooth function of ε/t^2 that equals $N_0(w)$ at 0, or 0 if $w = 0$, and the rest of the network is locally Lipschitz, so the output at tX is $G(X) + O(1/t^2)$ for some $G(X)$.

The residual network. Let $\bar{n} = N_0(XE)$ and $\beta(X) = \sum_\ell a_\ell(\bar{n}) W^\ell_{\text{write}}$, the sum of the outputs of all blocks at the normalized state $\bar{n}$. By induction over the blocks, $h_\ell = tXE + \sum_{k \leq \ell} a_k(\bar{n}) W^k_{\text{write}} + O(1/t)$: each hidden state is $tXE + O(1)$, so for large t it has no zero row, the next block reads $N(h_\ell) = \bar{n} + O(1/t)$, and, being locally Lipschitz, it adds its output at $\bar{n}$ up to $O(1/t)$. At the readout, on each row, with w the row of XE and b the row of $\beta(X)$, $N(tw + b + O(1/t)) = N_0\big(w + b/t + O(1/t^2)\big) + O(1/t^2) = N_0(w) + \mathrm{D}N_0(w)[b]/t + O(1/t^2)$. So the output at tX is

$$N_0(XE)D + \rho(X)/t + O(1/t^2), \qquad \rho(X) = \mathrm{D}N_0(XE)\big[\beta(X)\big]\, D,$$

with $\mathrm{D}N_0$ taken row by row.

Comparison. Let the two networks be equal. Both are continuous on the open set of inputs at which both are defined, so they agree on all of it. The skipless network is defined at tX if and only if it is defined at X, by the case $\varepsilon = 0$ above, and everywhere if $\varepsilon > 0$; the residual network is defined at tX for all large t. So if the skipless network is defined at X, their outputs at tX agree for all large t, and $\rho(X) = 0$. These inputs X are dense: for $\varepsilon = 0$ by the definition of equal networks, and for $\varepsilon > 0$ because E has rank d_{model}. As ρ is continuous on the inputs for which XE has no zero row, $\rho = 0$ there. As D has rank d_{model}, each row of $\beta(X)$ is then a multiple of the same row of XE. Here $\beta(X)$ depends on X only through $\bar{n}$, and as XE takes every value without zero rows, $\bar{n}$ takes every value whose rows have norm $\sqrt{d_{\text{model}}}$. So equality requires that for every such $\bar{n}$, each row of $\sum_\ell a_\ell(\bar{n}) W^\ell_{\text{write}}$ is a multiple of the same row of $\bar{n}$.

Measure zero. Fix all weights but the writes. The condition is linear in the writes, so the writes that satisfy it form a linear subspace. The subspace is proper if some $a_\ell(\bar{n})$ has a nonzero entry (r, j) at some $\bar{n}$: the write of block ℓ whose row j is not a multiple of row r of $\bar{n}$, with all other entries and all other writes zero, violates the condition. And a_1 vanishes for all $\bar{n}$ only if the read of block 1 has a zero column, which happens on a set of measure zero. The weights that satisfy the condition form a closed set, since every a_ℓ is continuous in the read and in $\bar{n}$, and by Fubini's theorem this set has measure zero. □

A.4 Skipless networks as limits of residual networks (Proposition 5.1)

Proposition 5.1 (Skipless networks as limits of residual networks)**.** *Let F be a skipless network whose blocks are locally Lipschitz.*

(i) Without normalization. *The network F is a limit of residual networks with blocks of the same*

kinds and widths, uniformly on bounded sets of inputs.
(ii) Under pre-RMSNorm with $\varepsilon = 0$. *The network F is a limit of residual networks with blocks of the same kinds and widths, uniformly on compact sets of inputs at which F is defined.*

Let F have the embedding E, the blocks B_ℓ, with reads W^ℓ_{read} and writes W^ℓ_{write}, and the readout D, and let $s_0 = XE$ and $s_\ell = B_\ell(s_{\ell-1})$ be its hidden states, so that $F(X) = s_L D$. For $t \geq 1$, let F_t be the residual network with the embedding E, the reads $t^{1-\ell}W^\ell_{\text{read}}$, the writes $t^\ell W^\ell_{\text{write}}$ and the readout $t^{-L}D$; under pre-RMSNorm with $\varepsilon = 0$, let F_t keep the embedding, the reads and the readout of F and have the writes $t^\ell W^\ell_{\text{write}}$. Let $r_0 = 0$ and

$$r_\ell \;=\; B_\ell(s_{\ell-1} + r_{\ell-1}) - B_\ell(s_{\ell-1}) \;+\; t^{-1}\,(s_{\ell-1} + r_{\ell-1}). \tag{11}$$

The proofs show that the hidden states of F_t are $t^\ell(s_\ell + r_\ell)$ and bound r_ℓ, which shows that F_t tends to F. Here ED may vanish: F_t has $ED = 0$ whenever F does. Norms are Frobenius norms for hidden states and operator norms for matrices.

Proof of Proposition 5.1(i). Block ℓ of F_t has its reads scaled by $t^{1-\ell}$ and its writes by t^ℓ, so it maps h to $t^\ell B_\ell(t^{1-\ell}h)$. We show by induction that its hidden states are $h_\ell = t^\ell(s_\ell + r_\ell)$, with $r_0 = 0$ and r_ℓ given by (11). For $\ell = 0$, $h_0 = XE = s_0$. If $h_{\ell-1} = t^{\ell-1}(s_{\ell-1} + r_{\ell-1})$, then

$$h_\ell \;=\; h_{\ell-1} + t^\ell B_\ell(t^{1-\ell}h_{\ell-1}) \;=\; t^\ell\big(B_\ell(s_{\ell-1} + r_{\ell-1}) + t^{-1}(s_{\ell-1} + r_{\ell-1})\big) \;=\; t^\ell(s_\ell + r_\ell).$$

Let $\mathcal{X}$ be a bounded set of inputs and $\bar{s}_\ell = \sup_{X\in\mathcal{X}}\|s_\ell\|$, which is finite because the blocks are continuous. Suppose that $\|r_{\ell-1}\| \leq C_{\ell-1}/t$ for all $X \in \mathcal{X}$ and $t \geq 1$, which holds for $\ell = 1$ with $C_0 = 0$. Then $s_{\ell-1}$ and $s_{\ell-1} + r_{\ell-1}$ lie in the ball of radius $\bar{s}_{\ell-1} + C_{\ell-1}$, on which B_ℓ is Lipschitz with some constant Lip_ℓ, since B_ℓ is locally Lipschitz. Hence

$$\|r_\ell\| \;\leq\; \text{Lip}_\ell\|r_{\ell-1}\| + t^{-1}\big(\bar{s}_{\ell-1} + \|r_{\ell-1}\|\big) \;\leq\; C_\ell/t, \qquad C_\ell = \text{Lip}_\ell C_{\ell-1} + \bar{s}_{\ell-1} + C_{\ell-1}.$$

Finally $F_t(X) = t^{-L}h_L D = F(X) + r_L D$, so $\|F_t(X) - F(X)\| \leq C_L\|D\|/t$ for all $X \in \mathcal{X}$ and $t \geq 1$. □

Proof of Proposition 5.1(ii). Let $\mathcal{X}$ be a compact set of inputs at which F is defined, that is, at which no row of XE or of a hidden state of F vanishes. Here N divides each row of h by its norm and multiplies it by $\sqrt{d_{\text{model}}}$, so $N(ch) = N(h)$ for $c > 0$, and $\|N(y') - N(y)\| \leq 2\sqrt{d_{\text{model}}}\,\|y' - y\|/\|y\|$ for nonzero rows y, y'. The hidden states of F are $s_0 = XE$ and $s_\ell = B_\ell(N(s_{\ell-1}))$, and $F(X) = N(s_L)D$. As no row of s_ℓ vanishes on $\mathcal{X}$, continuity and compactness give $\delta > 0$ below the norm of every row of every s_ℓ, $\ell \geq 1$, on $\mathcal{X}$. Since N removes the factor $t^{\ell-1}$, the hidden states of F_t are again $h_0 = s_0$ and $h_\ell = t^\ell(s_\ell + r_\ell)$, now with $r_0 = 0$ and

$$r_\ell \;=\; B_\ell\big(N(s_{\ell-1} + r_{\ell-1})\big) - B_\ell\big(N(s_{\ell-1})\big) \;+\; t^{-1}\,(s_{\ell-1} + r_{\ell-1}),$$

as long as no row of $s_{\ell-1} + r_{\ell-1}$ vanishes. The maps B_ℓ are Lipschitz on the bounded image of N, with some constants Lip_ℓ, and $\bar{s}_\ell = \sup_{X\in\mathcal{X}}\|s_\ell\|$ is finite. If every row of $r_{\ell-1}$ is shorter than δ, then $\|r_\ell\| \leq 2\sqrt{d_{\text{model}}}\,\text{Lip}_\ell\|r_{\ell-1}\|/\delta + t^{-1}(\bar{s}_{\ell-1} + \|r_{\ell-1}\|)$. Let $C_0 = 0$, $C_\ell = (2\sqrt{d_{\text{model}}}\,\text{Lip}_\ell/\delta + 1)C_{\ell-1} + \bar{s}_{\ell-1}$ and $t_0 = \max(1, \max_\ell 2C_\ell/\delta)$. For $t \geq t_0$, induction on ℓ gives $\|r_\ell\| \leq C_\ell/t \leq \delta/2$ on $\mathcal{X}$: every row of $r_{\ell-1}$ is then shorter than δ, so no row of $s_{\ell-1} + r_{\ell-1}$ vanishes and the inequality applies. Finally $\|F_t(X) - F(X)\| = \|(N(s_L + r_L) - N(s_L))D\| \leq 2\sqrt{d_{\text{model}}}\,\|D\|\,C_L/(\delta t)$ for all $X \in \mathcal{X}$ and $t \geq t_0$. □

Remark A.16 (The rate). The proofs give the rate $1/t$. Without normalization, for every bounded set $\mathcal{X}$ of inputs there is C such that $\|F_t(X) - F(X)\| \leq C/t$ for all $X \in \mathcal{X}$ and $t \geq 1$. Under pre-RMSNorm with $\varepsilon = 0$, the same holds for all $t \geq t_0$, for every compact set $\mathcal{X}$ of inputs at which F is defined, with t_0 depending on $\mathcal{X}$. The constants depend on $\mathcal{X}$ and on F.

B Universal approximation: width and biases

Lin and Jegelka (2018) show that residual ReLU networks whose blocks have a single hidden unit, $h \mapsto h + \mathrm{ReLU}(hw+c)\,v$ with $w \in \mathbb{R}^{d\times 1}$, $v \in \mathbb{R}^{1\times d}$ and a bias $c \in \mathbb{R}$, approximate every integrable function on $\mathbb{R}^d$ in L^1 as the depth grows, while ReLU networks without skips whose hidden layers have width at most the input dimension d cannot (their Proposition 2.1). The separation is confined to this narrow regime: with hidden layers of width $d+1$, ReLU networks without skips already approximate every continuous function uniformly on compact sets (Hanin and Sellke, 2017).

The construction also needs the biases, for the shifts and the cut-offs at constants of its blocks (their Proposition 3.2). Without biases, a ReLU network, with or without skips, is positively homogeneous of degree one, $F(\lambda x) = \lambda F(x)$ for $\lambda > 0$, since ReLU and the linear maps are. On every ray $\{tu : t > 0\}$ such a network is a multiple of t, and so is every limit of such networks, uniformly or in L^1 on a compact set, on almost every ray: the multiples of t form a closed space, and by Fubini's theorem L^1 convergence passes to almost every ray. In one dimension, for instance, every such network is $x \mapsto a\,\mathrm{ReLU}(x) - b\,\mathrm{ReLU}(-x)$, which stays at L^1 distance at least $(2-\sqrt{2})/3$ from x^2 on $[-1,1]$. On all of $\mathbb{R}^d$, no integrable function other than 0 can be approximated in L^1, since a positively homogeneous function other than 0 is not integrable.

C Blocks with biases

The main text treats blocks without biases, as in many current language models. With biases the answers change: every ReLU, GELU, ReLU^2 and gated block absorbs a linear term with more units, while $\tanh$ blocks still never do (Theorem C.1). Attention blocks never do when their scores ignore the positions of the tokens (Theorem C.1), and they do with rotary embeddings (Proposition C.6). To absorb its skip, almost every ReLU, GELU and gated block needs exactly d_{model} more units, and almost every ReLU^2 block between $2d_{\mathrm{model}}$ and $3d_{\mathrm{model}}$ (Theorem C.2). Table 2 collects these results.

Blocks with biases. A vanilla MLP block with biases computes $B_\theta(x) = f(xW_{\mathrm{read}} + b)\,W_{\mathrm{write}} + c$, with $\theta = (W_{\mathrm{read}}, b, W_{\mathrm{write}}, c)$, a unit bias $b \in \mathbb{R}^{1\times m}$ and an output bias $c \in \mathbb{R}^{1\times d_{\mathrm{model}}}$; unit i has the read w_i, the bias b_i and the write $\hat{w}_i$. In a gated block with biases, unit j has a gate read g_j (a column of W_{g}) and a gate bias b^{g}_j, a value read v_j (a column of W_{v}) and a value bias b^{v}_j, and a write $\hat{w}_j$, and

$$B_\theta(x) = \sum_{j=1}^{m} \sigma\left(xg_j + b^{\mathrm{g}}_j\right)\left(xv_j + b^{\mathrm{v}}_j\right)\hat{w}_j + c.$$

An attention block with biases adds a bias row to every query, key and value and to its output: with $\mathbf{1}$ the column of T ones,

$$B_\theta(h) = \sum_{i=1}^{H} P_i(h)\left(hW_{V,i} + \mathbf{1}b_{V,i}\right)W_{O,i} + \mathbf{1}b_O,$$

where $P_i(h)$ is the softmax, row by row, of $(hW_{Q,i}+\mathbf{1}b_{Q,i})(hW_{K,i}+\mathbf{1}b_{K,i})^\top/\sqrt{d_h}$, with or without a causal mask. These scores do not depend on the positions of the tokens; rotary embeddings, under which they do, follow in Proposition C.6. A block absorbs xM, for $M \neq 0$, if a block of the same kind with biases satisfies (2), or for attention, with the same mask, $hM + B_\theta(h) = B_{\theta'}(h)$ for all $h \in \mathbb{R}^{T\times d_{\mathrm{model}}}$.

Theorem C.1 (Exact absorption with biases)**.** *Let blocks have biases, and let $M \neq 0$.*
(i) *Every ReLU, GELU, ReLU^2, ReGLU, SwiGLU or GeGLU block absorbs xM with more hidden units.*
(ii) *No $\tanh$ block and no attention block as above absorbs xM, at any width.*
(iii) *For generic θ, no block $B_{\theta'}$ of these kinds and of the same width as B_θ satisfies (2).*

Part (iii) follows from (ii) for $\tanh$ and attention blocks, and from the proofs of Theorem C.2 for the others.

Theorem C.2 (Extra units with biases)**.** *Let blocks have biases, let M be invertible and let θ be generic.*
(i) *A ReGLU, SwiGLU or GeGLU block absorbs xM with d_{model} more hidden units, and not with fewer.*
(ii) *A ReLU or GELU block with at least d_{model} hidden units absorbs xM with d_{model} more hidden units, and not with fewer.*
(iii) *A ReLU^2 block with at least d_{model} hidden units absorbs xM with $3d_{\text{model}}$ more hidden units, and not with fewer than $2d_{\text{model}}$.*

Proven	**ReLU**	**GELU**	**tanh**	**ReLU^2**	**ReGLU, SwiGLU, GeGLU**	**Attention**	**Attention, rotary**
one block, same width	measure zero Thm. C.1(iii)		never Thm. C.1(ii)	measure zero Thm. C.1(iii)		never Thm. C.1(ii)	unknown
one block, more hidden units or heads	exactly d_{model} more Thm. C.2(ii)		never Thm. C.1(ii)	$2d_{\text{model}}$ to $3d_{\text{model}}$ more Thm. C.2(iii)	exactly d_{model} more Thm. C.2(i)	never Thm. C.1(ii)	$T^2\lceil d_{\text{model}}/d_h\rceil$ more suffice Prop. C.6

Table 2: What is proven for one block with biases. Never: for no weights. Measure zero: only for a null set of weights. Exactly d_{model} more: for almost every block, d_{model} more hidden units suffice and fewer do not. Scope in Section 2.

Remark C.3 (Constructions for every block). Every gated block absorbs xM with $\operatorname{rank} M$ more units, and for generic θ fewer do not (proof of Theorem C.2(i)). Every ReLU or GELU block absorbs xM with $2d_{\text{model}}$ more units, and with d_{model} when its read has rank d_{model}. Every ReLU^2 block absorbs xM with $4d_{\text{model}}$ more units, and with $3d_{\text{model}}$ when its read has rank d_{model}. Almost every ReLU^2 block with at least $d_{\text{model}}(d_{\text{model}}+3)/2$ units absorbs it with $d_{\text{model}}(d_{\text{model}}+3)/2$ more, which is fewer than $3d_{\text{model}}$ only for $d_{\text{model}}=2$, where it is five.

Remark C.4 (Bounded sets of inputs). GELU, SwiGLU, GeGLU, $\tanh$ and attention blocks are real-analytic in their input, so an equality on a nonempty open set of inputs holds everywhere, and Theorems C.1 and C.2 hold on every such set. ReLU, ReLU^2 and ReGLU blocks are not. On a ball $\mathcal{X}$ of inputs, a unit whose argument (for ReGLU, whose gate's argument) does not vanish on $\mathcal{X}$ is a polynomial there. If d_{model} units of a ReLU block with linearly independent reads, or $d_{\text{model}}(d_{\text{model}}+3)/2$ generic units of a ReLU^2 or ReGLU block, are active on all of $\mathcal{X}$, changing their writes and the output bias absorbs xM on $\mathcal{X}$ at the same width. So on $\mathcal{X}$, the entries "measure zero" of Table 2 fail for ReLU, ReLU^2 and ReGLU blocks, for an open set of blocks. The lower bounds below use inputs of unbounded size: rays $x=tu$ with $t\to\infty$, or the hyperplanes on which the arguments of the units vanish, wherever they lie.

Remark C.5 (Networks). Without normalization, absorbing the skip block by block, with $M=I$, shows that every residual network of ReLU^2, ReGLU, SwiGLU or GeGLU blocks with biases equals a skipless network with $4d_{\text{model}}$ (ReLU^2) or d_{model} (gated) more units per block; without biases, none with $ED\neq 0$ does (Theorem 4.1(i)). As without biases, every residual ReLU or GELU network whose reads have rank d_{model} equals a skipless one with d_{model} more units per block. A skipless network of $\tanh$ blocks with biases is bounded, so it equals no residual one with $ED\neq 0$. Networks with biases under pre-RMSNorm are not treated here.

Rotary embeddings. With rotary embeddings (Su et al., 2021), the head dimension d_h is even, and the query and the key of the token at position t are multiplied by $\mathcal{R}^{\,t}$, where the $d_h\times d_h$ matrix $\mathcal{R}$ rotates $d_h/2$ coordinate planes by fixed angles. With h_t row t of h, the score of the query at position t against the key at position s is then

$$\left(h_tW_{Q,i}+b_{Q,i}\right)\mathcal{R}^{\,t-s}\left(h_sW_{K,i}+b_{K,i}\right)^{\top}/\sqrt{d_h},$$

which depends on the two positions. Without biases, no such block absorbs a linear term (Theorem 3.4). With biases, a head whose queries and keys are its biases attends by position alone, and such heads add up to a skip.

Proposition C.6 (Rotary embeddings with biases)**.** *Let attention blocks have biases and rotary embeddings, with or without a causal mask, and let one of the rotary angles, ϑ, satisfy $k\vartheta \notin 2\pi\mathbb{Z}$ for $0 < k < T$. Then every attention block absorbs hM, for every M, with at most $T^2\lceil \operatorname{rank} M/d_h \rceil$ more heads.*

The first angle of Su et al. (2021) is $\vartheta = 1$, which satisfies the condition for every T. So without normalization, every residual network of such blocks equals a skipless network with $T^2\lceil d_{\text{model}}/d_h \rceil$ more heads per block.

Proof. Let $e \in \mathbb{R}^{1\times d_h}$ be a unit row in the plane that $\mathcal{R}$ rotates by ϑ, so that $e\,\mathcal{R}^k e^\top = \cos(k\vartheta)$. For $\tau \geq 0$, a head with $W_Q = W_K = 0$ and $b_Q = b_K = \tau\, d_h^{1/4} e$ has the scores $\tau^2 \cos\big((t-s)\vartheta\big)$ at every input. So its attention matrix P_τ does not depend on h: row t of P_τ is the softmax of these scores over the positions s that token t attends to. Among them, $s = t$ has the score τ^2, and every other s has a smaller one, as $\cos\big((t-s)\vartheta\big) < 1$ by the condition on ϑ. Hence $P_\tau \to I$ as $\tau \to \infty$. The matrices P_τ, $\tau \geq 0$, span a linear subspace of $\mathbb{R}^{T\times T}$. A linear subspace is closed, so it contains I:

$$I = \sum_{j=1}^{J} \lambda_j P_{\tau_j} \qquad \text{with } J \leq T^2 \text{ and real numbers } \lambda_j.$$

Write $M = M_1 + \cdots + M_n$ with $n = \lceil \operatorname{rank} M/d_h \rceil$ and every M_i of rank at most d_h. For each j and i, add to the block a head with the queries and keys above at $\tau = \tau_j$, no value bias, and the value–output product $\lambda_j M_i$. The new heads add $\sum_{j,i} \lambda_j P_{\tau_j}\, h\, M_i = hM$ to the block. □

Remark C.7 (The count and the sequence length). The count T^2 is not sharp: the rows of every P_τ sum to 1, which already gives $J \leq T^2 - T + 1$. The heads depend on T. Under a causal mask they also serve every shorter sequence, since the attention matrix of the first T' tokens is the top-left block of P_τ; without a mask, the same argument applied to the matrices P_τ of all lengths $2, \ldots, T$ at once gives heads that serve all these lengths. Some blocks absorb hM at the same width: with the value–output products $-\lambda_j M_i$, the heads of the proof alone form a block B_θ with $hM + B_\theta(h) = 0$.

Open questions. The exact number of extra units for ReLU^2 blocks lies between $2d_{\text{model}}$ and $3d_{\text{model}}$; for $d_{\text{model}} = 2$, five suffice once the block has five units (Remark C.3). For attention blocks with biases and rotary embeddings, the least number of heads, and whether almost every block fails to absorb a linear term at the same width, are open.

C.1 Constructions and blocks that never absorb

Let e_a be the a-th standard basis column of $\mathbb{R}^{d_{\text{model}}}$ and M_a the a-th row of M; then xM is the sum over a of $(xe_a)\, M_a$.

Proof of Theorem C.1(i). The constructions also give the upper bounds of Theorem C.2. Each construction keeps the units of B_θ, changes some of their writes and the output bias, and adds units.

Gated. A unit with gate read 0 and gate bias 1 computes $\sigma(1)\,(xv + b^{\text{v}})\,\hat{w}$, which is affine in x, and $\sigma(1) \neq 0$ for ReLU, SiLU and GELU. Write M as a sum of $\operatorname{rank} M$ matrices of rank one, each a column times a row. For each term, a unit with gate read 0, gate bias 1, value bias 0, the column as its value read and the row divided by $\sigma(1)$ as its write adds x times the term. So $\operatorname{rank} M \leq d_{\text{model}}$ units add xM.

ReLU and GELU. Both satisfy (3). Two new units with reads $\pm e_a$, biases 0 and writes $\pm M_a$ add $(xe_a)\, M_a$, so $2d_{\text{model}}$ new units add xM. If the read has rank d_{model}, choose d_{model} units with linearly independent reads, and let the rows γ_i, for these units i, solve $\sum_i w_i\gamma_i = M$. Give unit i the write $\hat{w}_i + \gamma_i$, and add a unit with read $-w_i$, bias $-b_i$ and write $-\gamma_i$. By (3) the two add $(xw_i + b_i)\,\gamma_i$, and subtracting $\sum_i b_i\gamma_i$ from the output bias leaves xM, with d_{model} new units.

ReLU^2. Let $r(z) = \max(z, 0)^2$, so that $r(z) + r(-z) = z^2$. Four new units compute $r\big(\pm(xe_a + 1)\big)$, with the write $M_a/4$, and $r\big(\pm(xe_a - 1)\big)$, with the write $-M_a/4$. Together they add $\big((xe_a + 1)^2 - (xe_a - 1)^2\big)M_a/4 = (xe_a)\, M_a$, so $4d_{\text{model}}$ new units add xM. If the read has rank d_{model}, take the units i and the rows γ_i as above, and let $z_i = xw_i + b_i$. Give unit i the write $\hat{w}_i + \gamma_i$

and add three units: one with read $-w_i$, bias $-b_i$ and write γ_i, and two with reads $\pm w_i$, biases $\pm(b_i - \frac{1}{2})$ and writes $-\gamma_i$. Together they add $\gamma_i\big(z_i^2 - (z_i - \frac{1}{2})^2\big) = (xw_i)\,\gamma_i + (b_i - \frac{1}{4})\,\gamma_i$, and the output bias absorbs the constants: $3d_{\text{model}}$ new units. Finally, for $d_{\text{model}}(d_{\text{model}}+3)/2$ units i of the block and new rows γ_i, give unit i the write $\hat{w}_i + \gamma_i$ and add a unit with read $-w_i$, bias $-b_i$ and write γ_i: these add $\sum_i \gamma_i z_i^2$. For each output coordinate o, $\sum_i \gamma_i z_i^2 = xM + \text{const}$ is a linear system of $d_{\text{model}}(d_{\text{model}}+3)/2$ equations, one for each coefficient of a polynomial of degree two other than the constant, in the $d_{\text{model}}(d_{\text{model}}+3)/2$ unknowns $\gamma_{i,o}$. For generic θ it has a solution. □

Proof of Theorem C.1(ii). $\tanh$. $B_{\theta'} - B_\theta$ is bounded, and xM is not.

Attention. Suppose that $hM + B_\theta(h) = B_{\theta'}(h)$ for all h, and let C and C' be the sums of the value–output products $W_{V,i}W_{O,i}$ of the heads of θ and of θ'. If all rows of h equal x, all keys of a head are equal, so each row of $P_i(h)$ is uniform on the tokens it sees, and all values equal $xW_{V,i} + b_{V,i}$. So every row of $B_\theta(h)$ is xC plus a constant row, and the equation gives $M = C' - C$. Now add $\epsilon\delta$ to the first row of this h, for a row δ, and look at the second row, which sees the first token with or without a causal mask. The second row of hM does not change. In the second row of $B_\theta(h)$, the derivatives of the attention weights at $\epsilon = 0$ sum to zero over the tokens and multiply equal values, so the derivative in ϵ at $\epsilon = 0$ is $\delta C/T_2$, with T_2 the number of tokens that the second token sees. Hence $\delta C = \delta C'$ for every δ, so $C = C'$ and $M = 0$, a contradiction. □

C.2 Lower bounds

For a unit j of a gated block with biases, let $\Delta_j = \big(b_j^{\text{g}} v_j + b_j^{\text{v}} g_j\big)\,\hat{w}_j$, a $d_{\text{model}} \times d_{\text{model}}$ matrix of rank at most one.

Lemma C.8 (Gated blocks with biases)**.** *Let ReGLU, SwiGLU or GeGLU blocks B_θ and $B_{\theta'}$ with biases satisfy (2), where $B_{\theta'}$ has n more units than B_θ. Let the gate reads of B_θ be nonzero and pairwise non-collinear, and its value reads and writes nonzero. Then for some set S of units of B_θ, the matrix $M + \sum_{i\in S} \Delta_i$ is a sum of at most n matrices of rank one.*

Proof. Outline. Along rays $x = tu$ with $t \to \infty$, every gate becomes a ReLU gate. The terms of order t^2 and t of (2) then split hyperplane by hyperplane, and those of order t express M through the units on one side of each hyperplane.

Expansion along rays. Group the units of both blocks with a nonzero gate read g by the hyperplane $\mathcal{H} = \{x : xg = 0\}$, choose a unit normal ν of $\mathcal{H}$, and write $g = s\eta\nu$ with $\eta > 0$ and a side $s = \pm 1$. Let $[\,\cdot\,]$ be 1 if the condition inside holds and 0 otherwise. For $x \notin \mathcal{H}$ and $t \to \infty$,

$$\sigma\big(txg + b^{\text{g}}\big) = t\eta\,\text{ReLU}(s\,x\nu) + b^{\text{g}}\,[s\,x\nu > 0] + o(1/t),$$

exactly for large t when σ is ReLU, and up to terms that decay exponentially in t when σ is SiLU or GELU. So the unit contributes

$$t^2\,\text{ReLU}(s\,x\nu)\,(xv)\,\eta\hat{w} + t\,\big(\text{ReLU}(s\,x\nu)\,\eta b^{\text{v}}\hat{w} + [s\,x\nu > 0]\,(xv)\,b^{\text{g}}\hat{w}\big) + O(1),$$

and a unit with gate read 0 contributes $t\,\sigma(b^{\text{g}})\,(xv)\,\hat{w} + O(1)$. Put tx into (2) and compare the coefficients of t^2 and of t at each x off all the hyperplanes. For a hyperplane $\mathcal{H}$ and a side $\pm$, let $A^\pm$, $a^\pm$ and $G^\pm$ be the sums of $\eta v\hat{w}$, $\eta b^{\text{v}}\hat{w}$ and $b^{\text{g}}v\hat{w}$ over the units of $B_{\theta'}$ on that side, minus the same sums over the units of B_θ, and let Z be the same difference for the sums of $\sigma(b^{\text{g}})\,v\hat{w}$ over the units with gate read 0. Then, with sums over the hyperplanes,

$$\begin{aligned} 0 &= \textstyle\sum_{\mathcal{H}} \big(\text{ReLU}(x\nu)\,xA^+ + \text{ReLU}(-x\nu)\,xA^-\big), \\ xM &= \textstyle\sum_{\mathcal{H}} \big(\text{ReLU}(x\nu)\,a^+ + \text{ReLU}(-x\nu)\,a^- + [x\nu > 0]\,xG^+ + [x\nu < 0]\,xG^-\big) + xZ. \end{aligned}$$

One hyperplane at a time. Near a point of $\mathcal{H}$ that lies on no other hyperplane, the terms of the other hyperplanes are smooth. In the second identity, continuity across $\mathcal{H}$ forces $yG^+ = yG^-$ for every row y with $y\nu = 0$, so $G^+ - G^- = \nu q$ for a row q. Then $[x\nu > 0]\,x\nu q = \text{ReLU}(x\nu)\,q$, so the

terms of $\mathcal{H}$ are $\mathrm{ReLU}(x\nu)\,(a^+ + q) + \mathrm{ReLU}(-x\nu)\,a^- + xG^-$. Since the identity has no kink at $\mathcal{H}$, $a^+ + q + a^- = 0$. The terms of $\mathcal{H}$ are then $x\Delta_{\mathcal{H}}$, with

$$\Delta_{\mathcal{H}} = G^- - \nu a^- = G^+ + \nu a^+,$$

and the second identity becomes $M = \sum_{\mathcal{H}} \Delta_{\mathcal{H}} + Z$. As $g = \pm\eta\nu$ on side $\pm$, $G^{\pm} \pm \nu a^{\pm}$ is the sum of the matrices Δ_j over the units of $B_{\theta'}$ on side $\pm$, minus the same sum over the units of B_θ. In the first identity, the same argument applied to $\mathrm{ReLU}(\pm x\nu) = (|x\nu| \pm x\nu)/2$ gives $A^+ + A^- = \nu p$ for a row p, and then $p = 0$, since $|x\nu|\,x\nu$ has a jump in its second derivative. So $A^+ + A^- = 0$.

Counting. Orient the hyperplane $\mathcal{H}_i$ of unit i of B_θ so that unit i lies on side $+$; no other unit of B_θ lies on $\mathcal{H}_i$. Let $J^{\pm}$ be the sets of units of $B_{\theta'}$ on side $\pm$ of $\mathcal{H}_i$. If both were empty, $A^+ + A^-$ would be $-\eta_i v_i \hat{w}_i$, where $g_i = \eta_i \nu$, which does not vanish; so $\mathcal{H}_i$ carries $|J^+| + |J^-| - 1 \geq 0$ extra units. If J^+ is empty, $\Delta_{\mathcal{H}_i} = -\Delta_i$. Otherwise $\Delta_{\mathcal{H}_i}$ is the sum of the Δ_j over J^-, at most $|J^-| \leq |J^+| + |J^-| - 1$ matrices of rank one. A hyperplane of no unit of B_θ carries $|J^+| + |J^-|$ units of $B_{\theta'}$, with $J^{\pm}$ defined in the same way, and $\Delta_{\mathcal{H}}$ is a sum of at most that many matrices of rank one. Each unit of $B_{\theta'}$ with gate read 0 adds one matrix of rank one to Z, and B_θ has no such unit. With S the units i for which J^+ is empty, $M + \sum_{i \in S} \Delta_i$ is therefore a sum of at most n matrices of rank one. □

Proof of Theorem C.2(i). The argument also gives Theorem C.1(iii) for gated blocks. The construction above gives d_{model} units. For each set S of units of B_θ, the determinant of $M + \sum_{i \in S} \Delta_i$ is a polynomial in θ, and it equals $\det M \neq 0$ when all biases vanish. So for generic θ it is nonzero for each of the finitely many sets S, the hypotheses of Lemma C.8 hold, and the lemma gives $n \geq d_{\mathrm{model}}$. For $M \neq 0$ of any rank, a minor of size $\mathrm{rank}\, M$ that does not vanish at M plays the role of the determinant, and gives $n \geq \mathrm{rank}\, M \geq 1$. □

Proof of Theorem C.2(ii). The argument also gives Theorem C.1(iii) for ReLU and GELU blocks. The construction above gives d_{model} units. For the lower bound, let $B_{\theta'}$ satisfy (2). ReLU changes by at most $|b|$ when its argument changes by b, and GELU differs from ReLU by at most $\kappa < 0.17$ (proof of Theorem 4.5). So for both activations and all $t > 0$,

$$\left|\sigma(tz + b) - t\,\mathrm{ReLU}(z)\right| \leq |b| + \kappa,$$

and $\sigma(tz + b)/t \to \mathrm{ReLU}(z)$ as $t \to \infty$. Put tx into (2), divide by t, and let $t \to \infty$: the ReLU blocks without biases that have the reads and writes of B_θ and $B_{\theta'}$ satisfy (2). For generic θ these reads and writes are generic, so $B_{\theta'}$ has at least d_{model} more units by Proposition A.11(ii), and for any $M \neq 0$ it is not of the same width, by Theorem 3.1(i). □

Proof of Theorem C.2(iii). The argument also gives Theorem C.1(iii) for ReLU^2 blocks. The construction above gives $3d_{\mathrm{model}}$ units. For the lower bound, let $B_{\theta'}$, with n more units than B_θ, satisfy (2), and let $r(z) = \max(z, 0)^2$.

Squares. Group the units of both blocks with a nonzero read by the hyperplane $\mathcal{H} = \{x : x\omega + \omega_0 = 0\}$ on which their argument vanishes, with a representative (ω, ω_0) and a side: a unit with read $\lambda\omega$ and bias $\lambda\omega_0$ computes $\lambda^2\, r\big(\pm(x\omega + \omega_0)\big)$ times its write, on side $\pm$ as $\lambda \gtrless 0$. Let $\mu^{\pm}$ be the sum of λ^2 times the write over the units of $B_{\theta'}$ on side $\pm$, minus the same sum over the units of B_θ. By (2), $B_{\theta'} - B_\theta$ is a polynomial. Near a point of $\mathcal{H}$ on no other hyperplane, the terms of $\mathcal{H}$ are $\mu^+ r(z) + \mu^- r(-z)$, with $z = x\omega + \omega_0$, whose second derivative in the direction of ω jumps across $\mathcal{H}$ by a nonzero multiple of $\mu^+ - \mu^-$. So $\mu^+ = \mu^- =: \mu_{\mathcal{H}}$, the terms of $\mathcal{H}$ are $\mu_{\mathcal{H}}\, z^2$, and

$$\textstyle\sum_{\mathcal{H}} \mu_{\mathcal{H}}\,(x\omega_{\mathcal{H}} + \omega_{0,\mathcal{H}})^2 = xM + \mathrm{const}, \tag{12}$$

with the representatives $(\omega_{\mathcal{H}}, \omega_{0,\mathcal{H}})$.

Costs. For generic θ the units of B_θ have nonzero reads and writes and lie on distinct hyperplanes. Take unit i itself, (w_i, b_i), as the representative of its hyperplane $\mathcal{H}_i$, on side $+$. Then the sums of λ^2 times the writes of the units of $B_{\theta'}$ on $\mathcal{H}_i$ are $\hat{w}_i + \mu_{\mathcal{H}_i}$ on side $+$ and $\mu_{\mathcal{H}_i}$ on side $-$, so $B_{\theta'}$ has at least one unit on $\mathcal{H}_i$, and at least two unless $\mu_{\mathcal{H}_i} \in \{0, -\hat{w}_i\}$. On a hyperplane of no unit of B_θ, it

has at least two units if $\mu_{\mathcal{H}} \neq 0$. Let S and U be the sets of units i of B_θ with $\mu_{\mathcal{H}_i} = -\hat{w}_i$ and with $\mu_{\mathcal{H}_i} \notin \{0, -\hat{w}_i\}$, and $\mathcal{N}$ the set of the other hyperplanes with $\mu_{\mathcal{H}} \neq 0$. Then $n \geq |U| + 2|\mathcal{N}|$.

Dimensions. Modulo constants, a polynomial of degree at most two is a pair of a quadratic form and a linear part, in a space of dimension $d_{\text{model}}(d_{\text{model}} + 3)/2$, in which $(x\omega + \omega_0)^2$ becomes $\varphi(\omega, \omega_0) = (\omega\omega^\top, 2\omega_0\omega)$. For each output coordinate o, with $M_{:,o}$ the column o of M, (12) reads

$$\textstyle\sum_{i \in U} \mu_{\mathcal{H}_i, o}\, \varphi(w_i, b_i) + \sum_{\mathcal{H} \in \mathcal{N}} \mu_{\mathcal{H}, o}\, \varphi(\omega_{\mathcal{H}}, \omega_{0,\mathcal{H}}) = \tau_o,$$

where $\tau_o = (0, M_{:,o}) + \sum_{i \in S} \hat{w}_{i,o}\, \varphi(w_i, b_i)$. So the vectors $\tau_1, \ldots, \tau_{d_{\text{model}}}$ and $\varphi(w_i, b_i)$, $i \in U$, lie in the span of $|U| + |\mathcal{N}|$ vectors.

Genericity. For each of the finitely many pairs of disjoint sets S and U of units of B_θ, these $d_{\text{model}} + |U|$ vectors span a space of dimension at least $\min\big(d_{\text{model}}(d_{\text{model}} + 3)/2, d_{\text{model}} + |U|\big)$ for generic θ. The condition is that some minor of the matrix with these columns, a polynomial in θ, does not vanish. It holds when the writes $\hat{w}_i$, $i \in S$, vanish and the reads are generic: then the $\tau_o = (0, M_{:,o})$ span all linear parts, as M is invertible, and for generic columns w_i, any $\min\big(|U|, d_{\text{model}}(d_{\text{model}} + 1)/2\big)$ of the forms $w_i w_i^\top$ are linearly independent.

Conclusion. So $|U| + |\mathcal{N}| \geq \min\big(d_{\text{model}}(d_{\text{model}} + 3)/2, d_{\text{model}} + |U|\big)$. If the minimum is $d_{\text{model}} + |U|$, then $|\mathcal{N}| \geq d_{\text{model}}$ and $n \geq 2d_{\text{model}}$. Otherwise $|U| + |\mathcal{N}| \geq d_{\text{model}}(d_{\text{model}} + 3)/2 \geq 2d_{\text{model}}$, and again $n \geq 2d_{\text{model}}$. For $M \neq 0$ of any rank, the same argument with $\operatorname{rank} M$ in place of d_{model} gives $n \geq 2$. □